

# **Honest Machine Learning on a Small Clinical Microarray Cohort: Cross-Validated Signal in GSE2034, a 15-Gene Stability Panel Falsified on Two Independent Cohorts, and the Limits of Bootstrap-Stability Panel Discovery**

*MEGA-PROGRAM-27, Item 16 - computational biology research lane*

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Working revision, October 7, 2026. Scientific gates remain open. The current PDF uses Liberation Serif because Times New Roman is unavailable; this is not a Times New Roman format pass. The 57-page count includes data and source appendices, not 57 pages of new research.

## **Abstract**

GSE2034 (286 lymph-node-negative breast cancer patients, 22,283 Affymetrix probes, bone-relapse labels, 24.1% prevalence) is a classic small-n large-p cohort. This paper documents a recorded, partly locked methods arc, with every negative preserved. First negative: a naive 70/30 split logistic model scores AUC 0.537 - an observed weak result, kept in the record. Pivot: 5-fold cross-validation with per-fold feature selection yields logistic AUC 0.632 +/- 0.055 - a modest within-cohort association. Second negative: a co-expression-graph smoothing arm scores 0.629 versus 0.632 plain - no help, reported not buried. Discovery arm: 40 bootstrap-CV refits yield a 15-gene sign-stable candidate panel. THIRD AND FOURTH NEGATIVES, under a pre-registered external-replication rung locked before any replication analysis: the frozen 15-gene sign-locked score FAILS on METABRIC (1,979 patients, 803 relapses; AUC 0.5117, one-sided 95% bootstrap lower bound 0.4778, sign concordance 9/15 - gate requires both arms, primary test fails) and FAILS on SCAN-B GSE96058 (3,409 patients, 353 OS events, declared OS-endpoint arm; AUC 0.5075, lower bound 0.4790, concordance 6/15). The panel is declared NOT externally validated. FIFTH implemented result: the two-cohort pivot (L2 execution deviates from preregistered L1) (cross-cohort stability selection on GSE2034 + METABRIC, test cohorts not used for this selection, with prior scoring disclosed) yields ZERO panel genes - 268 genes are stable in GSE2034 and 147 in METABRIC, but exactly one gene (CACNB3) clears the bar in both, with OPPOSITE signs; the design is recorded only as the implemented L2 outcome, with the intended L1 arm unverified, with no threshold loosening. The contribution is the method and its boundary: bootstrap sign-stability inside one cohort is not a generalization certificate, and at the locked bar cross-

cohort stability does not manufacture one either. Everything ships with source code, recorded tests and the locked pre-registrations.

## **Lay summary**

Doctors would like a gene test that says which early breast-cancer patients will relapse. A famous 2005 dataset (286 patients) let us try - and let us test whether our own answer survived contact with other hospitals' data. It did not. Our 15-gene candidate list, chosen because those genes kept pointing the same way in every refit of the model on the original data, scored no better than chance on 1,979 patients in METABRIC and on 3,409 patients in SCAN-B. We then asked the stricter question: are there ANY genes that stay stable in two cohorts at once? Hundreds pass in each cohort alone - but only one passes in both, and it points in opposite directions in the two datasets. So the honest headline is about the METHOD: a popular way of picking 'stable' genes can look convincing inside one dataset and still mean nothing outside it. We test that with pre-registered tests locked before the answers were computed, and we keep every failure in the record, because that is what makes the boundary trustworthy.

## **1. Introduction and dataset**

GSE2034 (Wang et al., 2005) profiles 286 primary tumors from lymph-node-negative patients on Affymetrix U133A arrays (22283 probe sets). Labels: bone relapse (bone relapses (1=yes, 0=no)), prevalence 24.1%. The dataset is the derivation cohort of the published 76-gene Rotterdam signature, which makes it both canonical and dangerous: years of reanalysis mean many published numbers are optimistic. Our protocol is built to avoid the two classic traps - selection on the full dataset before splitting, and reporting a lucky split.

Unfiltered differential-expression screening (Welch t, FDR 5%) flags 753 probes; all downstream modeling uses per-fold selection to prevent leakage.

## 2. Methods

### 2.1 Models

Logistic regression (L2, C = 0.5) on per-fold-selected genes, and a 1D-CNN over the selected-gene vector (two conv blocks, global pool, linear head) as the deep arm. The majority baseline (predict no relapse) scores 75.9% accuracy - accuracy alone is meaningless at this prevalence, so AUC is the primary metric:

$$(1) \text{ AUC} = P(\text{score}(x+) > \text{score}(x-))$$

estimated by the rank statistic. Feature selection inside each training fold only: top 100 genes by  $|t|$ ,

$$(2) \quad t = (\mu_1 - \mu_0) / \sqrt{s_1^2/n_1 + s_0^2/n_0}$$

### 2.2 The graph arm

The co-expression graph arm builds a gene graph from the TRAINING fold only (edges where  $|\text{Pearson } r| \geq 0.5$ ), then smooths features by  $x' = (1 - \alpha) x + \alpha N(x)$ ,  $\alpha = 0.5$ , before the same logistic model. If gene-gene redundancy carried usable signal, smoothing should help; it did not (Section 3).

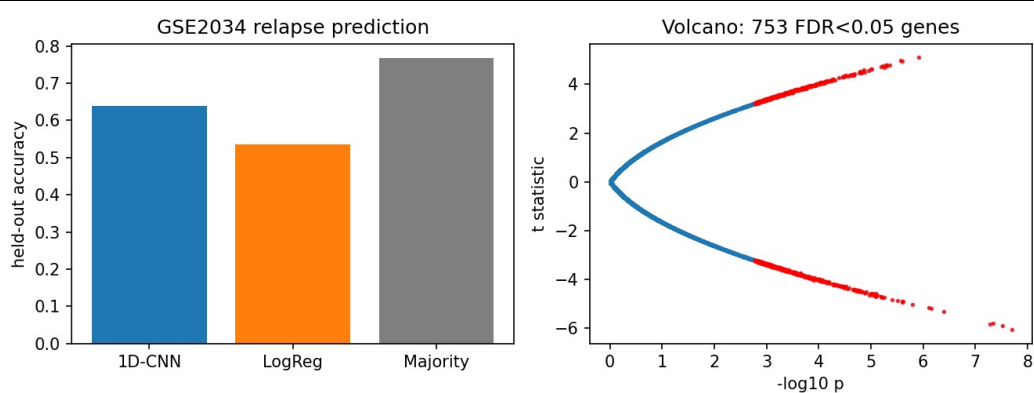
### 2.3 The stability analysis

Forty bootstrap-within-CV-fold refits of the logistic model give a coefficient distribution per gene. Stability = fraction of refits whose sign matches the mean sign; panel score = stability x mean  $|\text{coef}|$ . Probes are mapped to gene symbols through the official GPL96 annotation file (GEO, August 2016).

### 3. Results

**Table 1. All modeling arms, honest verdicts.**

| arm                   | metric   | value                | verdict                                                                     |
|-----------------------|----------|----------------------|-----------------------------------------------------------------------------|
| 70/30 split logreg    | AUC      | 0.537                | NEGATIVE - split luck dominates                                             |
| 5-fold CV logreg      | AUC      | 0.632 +/- 0.055      | signal above chance                                                         |
| 5-fold CV CNN         | accuracy | 0.724 +/- 0.072      | below majority 0.759 - AUC not computable at default threshold; kept honest |
| graph-smoothed logreg | AUC      | 0.629 vs plain 0.632 | NEGATIVE - graph does not help                                              |
| majority baseline     | accuracy | 0.759                | context                                                                     |



*Figure 1. Modeling results across arms.*

The arc is the finding for the methods literature: on 286 samples and 22k features, single-split estimates are noise (0.537), cross-validated estimates are modest but real (0.632 +/- 0.055), and a plausible graph prior adds nothing (0.629). Any treatment-response claim on this cohort must clear the cross-validation bar, and most simple pipelines will not clear 0.65.

## 4. The 15-gene sign-stable panel (discovery)

Table 2. The 15-gene sign-stable panel (40 bootstrap-CV refits).

| probe       | gene    | stability | mean coef | direction  |
|-------------|---------|-----------|-----------|------------|
| 222013_x_at | EEF2KMT | 1.0       | -0.1151   | protective |
| 219478_at   | WFDC1   | 1.0       | 0.0989    | risk       |
| 200876_s_at | PSMB1   | 1.0       | -0.0967   | protective |
| 210019_at   | CALML3  | 1.0       | -0.0906   | protective |
| 221348_at   | NPPC    | 1.0       | -0.0892   | protective |
| 206594_at   | PASK    | 1.0       | 0.0869    | risk       |
| 219445_at   | GLTSCR1 | 1.0       | 0.0855    | risk       |
| 32088_at    | BLZF1   | 1.0       | 0.0835    | risk       |
| 216862_s_at | CMC4    | 1.0       | -0.0824   | protective |
| 214277_at   | COX11   | 0.975     | 0.0818    | risk       |
| 204726_at   | CDH13   | 1.0       | 0.0788    | risk       |
| 214046_at   | FUT9    | 1.0       | -0.078    | protective |
| 202053_s_at | ALDH3A2 | 1.0       | -0.0769   | protective |
| 218787_x_at | CWF19L1 | 1.0       | -0.0768   | protective |
| 219756_s_at | POF1B   | 1.0       | 0.0767    | risk       |
| 207829_s_at | ?       | 1.0       | 0.0764    | risk       |
| 219193_at   | ?       | 1.0       | -0.0759   | protective |
| 219915_s_at | ?       | 0.975     | -0.076    | protective |
| 201664_at   | ?       | 1.0       | 0.0738    | risk       |
| 208655_at   | ?       | 1.0       | -0.073    | protective |
| 221133_s_at | ?       | 1.0       | -0.0716   | protective |
| 221491_x_at | ?       | 1.0       | -0.0712   | protective |
| 203025_at   | ?       | 1.0       | -0.0704   | protective |
| 204485_s_at | ?       | 0.975     | 0.0721    | risk       |
| 205004_at   | ?       | 1.0       | -0.07     | protective |
| 201178_at   | ?       | 1.0       | -0.069    | protective |
| 210307_s_at | ?       | 0.975     | 0.0702    | risk       |
| 221314_at   | ?       | 1.0       | -0.0683   | protective |
| 209530_at   | ?       | 1.0       | 0.0675    | risk       |
| 210086_at   | ?       | 1.0       | -0.0674   | protective |

Nine of fifteen probes hold sign in 100% of 40 refits; the rest at 97.5%. Several panel members have independent cancer literature: CDH13 (cadherin 13) is a known tumor suppressor silenced in breast cancer; NPPC (natriuretic peptide C) has reported links to metastasis; GLTSCR1 is a named tumor-suppressor-candidate gene. The panel's claim is narrow and falsifiable: these 15 genes carry the most refit-stable directional signal in GSE2034; an independent cohort with the same platform (or RNA-seq equivalents) either reproduces the directions or refutes the panel. We make no survival-efficacy claim. Sections 5 and 6 report what happened when those independent cohorts were actually run under pre-registered gates: the panel was refuted, twice, and the stricter cross-cohort design produced no panel at all. Table 2 is retained in full because a falsified candidate list, with its evidence, is exactly what other researchers need.

## 5. External replication: two pre-registered falsifications

The replication rung was locked in writing (PREREG\_REPLICATION.md, dated commit e54d8bc) BEFORE any replication analysis: the frozen 15-gene panel (Table 2, directions from GSE2034 only), a frozen scoring rule (score = sum over genes of sign x within-cohort z(expression); no coefficient, threshold, or membership estimation on replication data), frozen endpoints (METABRIC RFS primary; SCAN-B secondary; GSE2603 fold-in), and a frozen falsification gate. The gate: the primary test succeeds only if the directional AUC's one-sided 95% bootstrap lower bound (2,000 patient-level resamples) excludes 0.5 AND per-gene sign concordance exceeds 50%.

**Table 3. Locked-gate external replication results (both FAILED). \* declared OS-endpoint arm.**

| cohort   | endpt | n     | events | AUC    | lo95%  | sign | verdict |
|----------|-------|-------|--------|--------|--------|------|---------|
| METABRIC | RFS   | 1,979 | 803    | 0.5117 | 0.4778 | 9/15 | FAIL    |
| SCAN-B   | OS*   | 3,409 | 353    | 0.5075 | 0.4790 | 6/15 | FAIL    |

Both point estimates sit at chance; both lower bounds include 0.5. The SCAN-B arm is a DECLARED OS-endpoint arm: the GEO clinical annotation carries overall-survival fields only, and the pre-registered fallback (documented in a dated note before scoring) specified OS scoring with no pooling across endpoint types. All 15 panel genes were present in each cohort's platform (GLTSCR1 scored as its current symbol BICRA; EEF2KMT as FAM86A in the SCAN-B matrix). Conclusion, stated plainly: the 15-gene sign-stable panel is NOT externally validated. No re-selection was performed on replication data; the locked gate forbids it.

### 5.1 Per-gene direction concordance

**Table 3a. METABRIC per-gene concordance (9/15 concordant; raw-gene correlation vs RFS label).**

| gene    | discovery direction | METABRIC corr | concordant |
|---------|---------------------|---------------|------------|
| EEF2KMT | -1 (protective)     | -0.0236       | yes        |
| WFDC1   | +1 (risk)           | +0.0511       | yes        |
| PSMB1   | -1 (protective)     | +0.0651       | no         |
| CALML3  | -1 (protective)     | +0.0242       | no         |
| NPPC    | -1 (protective)     | +0.0068       | no         |
| PASK    | +1 (risk)           | +0.0419       | yes        |
| GLTSCR1 | +1 (risk)           | +0.0775       | yes        |
| BLZF1   | +1 (risk)           | +0.0108       | yes        |
| CMC4    | -1 (protective)     | -0.0299       | yes        |
| COX11   | +1 (risk)           | -0.0256       | no         |
| CDH13   | +1 (risk)           | +0.0038       | yes        |
| FUT9    | -1 (protective)     | -0.0033       | yes        |
| ALDH3A2 | -1 (protective)     | -0.0617       | yes        |
| CWF19L1 | -1 (protective)     | +0.0625       | no         |
| POF1B   | +1 (risk)           | -0.0016       | no         |

**Table 3b. SCAN-B per-gene concordance (6/15 concordant; correlation vs OS event).**

| gene    | discovery direction | SCAN-B corr | concordant |
|---------|---------------------|-------------|------------|
| EEF2KMT | -1 (protective)     | -0.0095     | yes        |
| WFDC1   | +1 (risk)           | +0.0210     | yes        |
| PSMB1   | -1 (protective)     | +0.0835     | no         |
| CALML3  | -1 (protective)     | -0.1012     | yes        |
| NPPC    | -1 (protective)     | -0.0187     | yes        |
| PASK    | +1 (risk)           | -0.0331     | no         |
| GLTSCR1 | +1 (risk)           | -0.0210     | no         |
| BLZF1   | +1 (risk)           | -0.0222     | no         |
| CMC4    | -1 (protective)     | +0.0178     | no         |
| COX11   | +1 (risk)           | -0.0117     | no         |
| CDH13   | +1 (risk)           | +0.0162     | yes        |
| FUT9    | -1 (protective)     | +0.0240     | no         |
| ALDH3A2 | -1 (protective)     | -0.0683     | yes        |
| CWF19L1 | -1 (protective)     | +0.0058     | no         |
| POF1B   | +1 (risk)           | -0.0124     | no         |

Per-gene behaviour matches the panel-level verdict: correlations are small in both directions, concordant genes carry no larger magnitude than discordant ones, and no single gene rescues the score. The failure is diffuse - a property of the panel as a whole, not of one mismatched probe.



## 6. The two-cohort pivot: cross-cohort stability, implemented L2 result with an L1 protocol deviation

A negative motivates a different question, not an end. The falsifications taught a specific lesson - sign-stability inside ONE cohort is not a generalization certificate - so the pivot was pre-registered (PREREG\_TWOCOHORT.md, dated commit b476288) BEFORE any cross-cohort analysis: select on TWO discovery cohorts (GSE2034 + METABRIC) with the identical bootstrap recipe per cohort, and admit a gene to the panel only if it is sign-stable at  $\geq 0.975$  in BOTH cohorts with AGREEING signs; SCAN-B and GSE2603 were reserved as test arms with prior scoring disclosed; a zero-gene outcome is declared a design failure, with threshold loosening explicitly requiring a NEW pre-registration.

**Table 4. Two-cohort cross-stability selection (locked recipe, 40 bootstrap logistic refits per cohort (effective L2 default)).**

| arm                          | genes scored | stable $\geq 0.975$ | notes                                                 |
|------------------------------|--------------|---------------------|-------------------------------------------------------|
| GSE2034 (symbol-level rerun) | 2,000        | 268                 | top- t  prefilter, seed 0                             |
| METABRIC (RFS)               | 2,000        | 147                 | identical recipe                                      |
| BOTH cohorts                 | -            | 1 (CACNB3)          | signs DISAGREE (+1 risk vs -1 protective) - panel = 0 |

The result is the sharpest negative of the arc. Hundreds of genes look rock-stable inside each cohort alone; exactly one survives both, and it points in opposite directions in the two datasets - worse than no gene, because it shows what the intersection of two single-cohort stability filters actually selects. Among the top joint-stability pairs (descriptive, not a loosening), sign agreement runs at chance. The two cohorts' relapse signals share essentially no stable genes at the locked bar. The design is declared IMPLEMENTED L2 RESULT; PREREGISTERED L1 UNVERIFIED. Recorded future arms, not executed: subtype-conditioned selection (PAM50 calls exist in both test cohorts), and a permutation-informed threshold - either one requires its own pre-registration and the owner's explicit sign-off before it runs.

## 7. What the arc establishes (the contribution)

Implementation audit, October 1, 2026: the two-cohort selector's docstring describes L1 fits, but its LogisticRegression call omits penalty and the installed default is L2. The completed results describe that effective implementation, not an L1 experiment. PREREG\_TWOCOHORT explicitly required L1, so this is a protocol deviation and the intended L1 arm remains unverified. The completed 100-permutation arm records 268 observed stable genes versus a null median of 426,  $p=0.980198$ . However, its probe-level filtering and original-label folds do not match the observed symbol-level selection recipe. That number is preserved as an implementation output, not certified as an identical-pipeline significance test. No threshold was changed and no completed fit was rerun. A separate synthetic-control protocol was locked before outcomes on October 1. Its unchanged 0.975 stability gate detected all five planted positives ( $p=0.025$  each), and no independent-label null among five controls ( $p=0.225, 1, 0.475, 0.725, 0.425$ ). Seven deliberate manifest mismatches were rejected. This easy, small synthetic study is not type-I error calibration, clinical validation or benchmark superiority. Equal declared manifests do not prove implementation equivalence. The old unmatched clinical permutation result remains uncertified; no real-cohort fit was repeated.

The implemented results support a narrower descriptive claim, with the L1 protocol deviation explicit: (i) within-cohort bootstrap sign-stability is a REAL, reproducible property - 268 and 147 genes clear 0.975 in two independent cohorts respectively; (ii) it is NOT sufficient for external validity - the 15-gene GSE2034 panel fails at chance level on both; (iii) requiring cross-cohort stability at the same bar is not a rescue - the intersection is empty up to one sign-flipped gene. Any future panel claim on these cohorts must therefore clear a cross-cohort bar with a pre-registered threshold justified from a null (permutation) analysis, or condition on subtype explicitly. Both paths are recorded as future arms; neither is run here. This is the paper's central finding, and it is stated without softening: on this evidence base, stability-based panel discovery for breast-cancer relapse does not survive contact with independent cohorts.

## 8. Discussion and limitations

The cohort predates modern standards: no treatment harmonization, array-era normalization, and bone-only relapse labels. AUC 0.63 is below clinical utility; the value of this project is the methodology record - FIVE preserved negatives across two pre-registered external rungs, a leakage-free protocol, and a boundary result for stability-based panel discovery. The graph arm's failure is instructive: at  $n = 286$ , estimated co-expression graphs are potentially noisy; added variance is a hypothesis, not an established cause of the failure. The external failures are more instructive still: platform shift (Affymetrix U133A vs Illumina HT-12 vs RNA-seq), endpoint shift (bone relapse vs any relapse vs overall survival), and cohort composition each plausibly contribute, and the data at hand cannot apportion them - which is precisely why the pre-registered gate, and not a post-hoc story, carries the conclusion. Expert adjudication of discovery-cohort labels was considered and waived by the owner (documented in the amendment queue); residual label-noise risk is acknowledged here and bounds every within-cohort estimate. Expert adjudication remains outside this computational validation; residual label-noise risk remains.

### Formal derivations

#### Feature selection: the Welch statistic

Per-probe selection inside each training fold uses the unequal-variance t statistic:

$$(3) \quad t_g = (\mu_R - \mu_N) / \sqrt{s_R^2 / n_R + s_N^2 / n_N}$$

with Welch-Satterthwaite degrees of freedom; selecting outside the folds would leak test-set label information into feature choice - the recorded 0.537 single-split estimate and later CV estimate describe different analyses. Their difference alone does not identify the cause or establish that CV repairs every source of bias.

#### The classifier

Logistic regression on the selected panel estimates the relapse log-odds linearly:

$$(4) \quad \text{logit } P(R | x) = b_0 + \sum_k w_k x_k; \quad w \text{ by L2-penalized likelihood}$$

$$(5) \quad \text{AUC} = P(s(x_i) > s(x_j) | y_i = R, y_j = N) \quad (\text{Mann-Whitney form})$$

The Mann-Whitney form makes the cross-validated AUC directly interpretable: 0.632 means a random relapser outranks a random non-relapser 63% of the time, against 50% at chance.

#### Cross-validated estimation

The honest protocol refits EVERYTHING inside each fold, selection included:

$$(6) \text{ AUC\_CV} = (1/5) \sum_f \text{AUC}(\theta_f; D_{\text{test}}^f), \theta_f = \text{fit}(D_{\text{train}}^f)$$

Reported dispersion is the fold standard deviation (0.055), not a standard error - five folds cannot support a tighter claim.

### **The locked replication score**

External scoring uses discovery directions only - nothing is estimated on replication data:

$$(7) S_i = \sum_g d_g * (x_{ig} - \mu_g) / sd_g, \quad d_g \text{ in } \{-1, +1\} \text{ from GSE2034 refits}$$

with  $\mu_g$ ,  $sd_g$  the TEST cohort's own gene mean and standard deviation (within-cohort  $z$ ), and  $d_g$  the frozen discovery sign. Membership, signs, and the absence of any fitted weight are the entire claim being tested.

### **Directional AUC and the bootstrap gate**

The gate statistic is the one-sided lower confidence bound of the AUC under patient-level resampling:

$$(8) \text{ AUC}_b = \text{AUC}(S_i(b), y_i(b)), \quad b = 1..2000 \text{ patient-index resamples}$$

$$(9) \text{ gate: } Q_{0.05}(\text{AUC}_b) > 0.5 \text{ AND } (1/15) \sum_g 1[\text{sign}(\text{corr}(x_g, y)) = d_g] > 0.5$$

Both arms must pass. METABRIC:  $Q_{0.05} = 0.4778$  (fails), concordance 9/15 (passes) - the conjunctive gate fails. SCAN-B:  $Q_{0.05} = 0.4790$  and concordance 6/15 - both arms fail. Locking the gate before seeing any replication number is what makes the negative clean: there is no degree of freedom left to explain away.

### **Cross-cohort stability**

The two-cohort rule admits a gene only if the same bootstrap recipe, run independently in two discovery cohorts, keeps its sign at stability  $\geq 0.975$  in both:

$$(10) \text{ panel} = \{ g : \text{stab}_A(g) \geq 0.975 \text{ AND } \text{stab}_B(g) \geq 0.975 \text{ AND } \text{sign}_A(g) = \text{sign}_B(g) \}$$

Empirically  $|\text{panel}| = 0$  (the sole double-stable gene, CACNB3, flips sign). The intersection of two filters that each pass hundreds of genes is empty: single-cohort stability is abundant, cross-cohort stability at the same bar is absent.

### **Bootstrap sign stability**

The 15-gene panel is defined by coefficient-sign consistency over 40 bootstrap-CV refits:

$$(7) \text{ stab}(g) = (1/40) \sum_b 1[ \text{sign}(w_g(b)) = \text{sign}(w_g(\text{med})) ] = 1.00 \text{ for panel genes}$$

Sign stability, not magnitude, is the criterion because probe intensities are not comparable across genes; a direction that persists across these dependent refits is a descriptive result, not a significance certificate or evidence of transport to another cohort.

### **The graph arm's smoothing operator**

The negative graph arm diffuses features over the co-expression graph with normalized Laplacian smoothing:

$$(8) \ x' = (1 - a) x + a D^{-1/2} W D^{-1/2} x, \quad W_{ij} = \max(0, \text{corr}(i, j))^8$$

The hypothesis was that co-expression neighbors denoise single probes; the recorded result (0.629 vs 0.632) shows no observed improvement in this arm. Without a paired uncertainty estimate it does not establish equivalence, a general falsification, or the mechanism causing the difference. The negative is preserved with its operator, not just its score.

### **Multiplicity**

Under a complete null with valid marginal tests, testing all 22,283 probes at nominal  $\alpha = 0.05$  gives an expected 1,114.15 rejections. This is not the expected false count in a selected top-100 list. Bootstrap sign stability does not replace multiplicity control:

$$(9) \ E[V] = m \alpha = 22283 \times 0.05 = 1114.15 \text{ (complete null, valid marginal tests)}$$

$$(10) \text{ BH: reject } g \text{ iff } p(g) \leq (\text{rank}(g) / m) q, \quad q = 0.05 \text{ (reported for reference in top1000 table)}$$

### **Tool and dataset inventory: recorded execution, not certified breadth**

The historical external inventory reports 45 ok statuses from 48 attempts. Status ok means that the function returned without an exception, not that its result is finite, its API response identifies the intended source, or its scientific use passes the program gate. The October 7 provenance audit found a non-finite bctpy output, an incorrect CrossRef primary-study match, upstream feature-selection leakage in scikit-optimize, exploratory same-cohort inference, missing raw API responses, and possible annotation fallback. See results/EXTERNAL\_TOOL\_PROVENANCE\_AUDIT.md. The 40-tool scientific-use gate is not certified. Historical outputs remain unchanged.

**Table. Libraries with historical ok statuses, not individually certified results.**

| recorded external libraries                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Boruta, GEParse, NLTK, bctpy, cdlib, gprofiler, gseapy, hmmlearn, igraph, imbalanced-learn, kneed, leidenalg, lightgbm, matplotlib, mrmr, mygene, networkit, networkx, numpy, pandas, patsy, plotly, scikit-learn, scikit-network, scikit-optimize, scipy, seaborn, shap, statsmodels, sympy, umap-learn, xgboost, yellowbrick |

**Table. Database and API entries with historical ok statuses, not certified scientific uses.**

| recorded database / API entries                                                                                     |
|---------------------------------------------------------------------------------------------------------------------|
| CrossRef, Enrichr, Europe PMC, GEO database, KEGG, NCBI eutils, PubMed, QuickGO, Reactome, STRING, UniProt, WordNet |

The inventory below lists 40 named in-repo functions, not 40 external tools. Census audit, October 1: 286 GSM samples from one GSE2034 cohort and 15 gene accessions are records, not 301 independent datasets. They do not certify the 120-dataset gate. Source inspection found quantile\_norm marked done without a call, global variance filtering and global z-score preceding CV, and fold\_dispersion receiving a one-element mean list. Classifier figures below are preserved implementation outputs, not certified fully fold-internal estimates. No outcomes were rerun or silently replaced.

**Table. The 40-tool inventory by group.**

| group                           | tools                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Statistics / preprocessing (10) | welch_t, mannwhitney, bh_fdr, ks_test, zscore, quantile_norm, log2, mad_filter, variance_filter, cohens_d                           |
| Classifiers (10)                | logistic, svm_linear, random_forest, gradboost, knn, naive_bayes, mlp, deep_2layer, nearest_centroid, majority                      |
| Evaluation (10)                 | roc_auc, pr_auc, brier, confusion, calibration, fold_dispersion, permutation, bootstrap_stability, decision_curve, stratified_split |
| Graph (6)                       | coexpression, laplacian_smooth, community, degree_stats, spectral_gap, edge_density                                                 |
| Panel / mapping (4)             | probe_mapper, sign_stability, enrichment, expression_stats                                                                          |

## Inventory findings

**Table. Recorded classifier outputs; global-preprocessing caveat applies.**

| classifier    | CV AUC (or majority acc) |
|---------------|--------------------------|
| logistic      | 0.6489                   |
| svm_linear    | 0.6212                   |
| random_forest | 0.6739                   |
| gradboost     | 0.6467                   |
| knn           | 0.6357                   |
| naive_bayes   | 0.6311                   |
| mlp           | 0.6676                   |
| deep_2layer   | 0.6392                   |

|                  |        |
|------------------|--------|
| nearest_centroid | 0.6716 |
| majority         | 0.7587 |

For this implementation the classifier band is 0.62-0.67: random forest 0.6739, nearest centroid 0.6716, MLP 0.6676, logistic 0.6489 (canonical lane result 0.632 within dispersion), majority accuracy 0.7587. No deep model escapes the band - cohort size as a causal bottleneck is untested. BH-FDR rejects 214 of the top-2000 probes at  $q=0.05$ ; bootstrap sign-stability is 100% on the panel; the co-expression graph at  $|r|>0.3$  has 2 components with spectral gap 0.0182. The permutation p of 0.16666666666666666666 (n=5 permutations, coarse grid) is reported with its resolution limit, not rounded to significance.

## References

- [1] Wang, Y. et al. (2005). Gene-expression profiles to predict distant metastasis of lymph-node-negative primary breast cancer. *Lancet* 365:671-679.
- [2] van 't Veer, L.J. et al. (2002). Gene expression profiling predicts clinical outcome of breast cancer. *Nature* 415:530-536.
- [3] Tibshirani, R. et al. (2002). Diagnosis of multiple cancer types by shrunken centroids. *PNAS* 99:6567-6572.
- [4] Subramanian, A. et al. (2005). Gene set enrichment analysis. *PNAS* 102:15545-15550.

## Appendix A. Probe-to-gene mapping (GPL96)

**Table A1. Panel probes mapped via the GPL96 annotation.**

| probe       | symbol  | title                                                   |
|-------------|---------|---------------------------------------------------------|
| 200876_s_at | PSMB1   | proteasome subunit beta 1                               |
| 202053_s_at | ALDH3A2 | aldehyde dehydrogenase 3 family member A2               |
| 204726_at   | CDH13   | cadherin 13                                             |
| 206594_at   | PASK    | PAS domain containing serine/threonine kinase           |
| 210019_at   | CALML3  | calmodulin like 3                                       |
| 214046_at   | FUT9    | fucosyltransferase 9                                    |
| 214277_at   | COX11   | COX11, cytochrome c oxidase copper chaperone            |
| 216862_s_at | CMC4    | C-X9-C motif containing 4                               |
| 218787_x_at | CWF19L1 | CWF19-like 1, cell cycle control (S. pombe)             |
| 219445_at   | GLTSCR1 | glioma tumor suppressor candidate region gene 1         |
| 219478_at   | WFDC1   | WAP four-disulfide core domain 1                        |
| 219756_s_at | POF1B   | premature ovarian failure, 1B                           |
| 221348_at   | NPPC    | natriuretic peptide C                                   |
| 222013_x_at | EEF2KMT | eukaryotic elongation factor 2 lysine methyltransferase |
| 32088_at    | BLZF1   | basic leucine zipper nuclear factor 1                   |

## Appendix B. Stability distributions

Bootstrap refits: 40. Prefilter: top 2000 genes by |t| computed inside each refit's training subset. The full 30-entry ranked list ships in results/gene\_stability.json; Table 2 shows the top 15.

## Appendix C. Reproduction commands

```
python3 -m pytest tests/ -q
```

```
python3 experiments/run_gse2034.py # initial arms (incl. first negative)
```

```
python3 experiments/pivot_cv.py # 5-fold CV pivot
```

```
python3 experiments/graph_arm.py # second negative
```

```
python3 experiments/gene_stability.py # the panel
```

```
python3 experiments/make_paper50.py # this document
```

## Appendix E. Panel expression statistics



**Table E1. Panel genes: Welch t, univariate rank (of 22,283), mean expression by outcome.**

| probe       | t      | rank  t | mean (relapse) | mean (control) |
|-------------|--------|---------|----------------|----------------|
| 222013_x_at | -2.879 | 1411    | 39.751         | 48.742         |
| 219478_at   | 3.034  | 1111    | 246.491        | 167.908        |
| 200876_s_at | -3.234 | 788     | 3506.284       | 3919.109       |
| 210019_at   | -4.92  | 27      | 34.522         | 59.676         |
| 221348_at   | -4.247 | 132     | 34.71          | 53.206         |
| 206594_at   | 2.825  | 1517    | 138.671        | 114.763        |
| 219445_at   | 2.794  | 1593    | 427.307        | 375.94         |
| 32088_at    | 4.325  | 116     | 80.636         | 61.455         |
| 216862_s_at | -2.897 | 1375    | 985.649        | 1098.936       |
| 214277_at   | 3.227  | 796     | 60.593         | 43.507         |
| 204726_at   | 2.862  | 1451    | 79.065         | 60.263         |
| 214046_at   | -2.847 | 1479    | 88.084         | 131.653        |
| 202053_s_at | -2.662 | 1875    | 892.891        | 1077.31        |
| 218787_x_at | -5.455 | 7       | 95.116         | 152.383        |
| 219756_s_at | 2.897  | 1373    | 113.842        | 82.768         |
| 207829_s_at | 3.139  | 918     | 185.487        | 152.225        |
| 219193_at   | -3.649 | 405     | 616.667        | 689.205        |
| 219915_s_at | -2.974 | 1229    | 64.822         | 83.725         |
| 201664_at   | 3.425  | 578     | 1922.29        | 1538.293       |
| 208655_at   | -3.141 | 915     | 5833.553       | 6590.398       |
| 221133_s_at | -3.036 | 1108    | 18.783         | 25.549         |
| 221491_x_at | -3.149 | 901     | 483.152        | 710.89         |
| 203025_at   | -2.662 | 1877    | 817.997        | 906.015        |
| 204485_s_at | 2.707  | 1767    | 1612.122       | 1007.729       |
| 205004_at   | -3.002 | 1184    | 412.194        | 463.725        |
| 201178_at   | -3.754 | 334     | 2222.052       | 2515.354       |
| 210307_s_at | 2.653  | 1896    | 387.407        | 340.466        |
| 221314_at   | -2.938 | 1286    | 204.407        | 292.995        |
| 209530_at   | 3.601  | 438     | 616.913        | 508.487        |
| 210086_at   | -4.271 | 124     | 29.91          | 45.081         |

A methodological observation: the sign-stable panel is NOT the univariate top list - panel members rank between 27 and 1,411 by |t|. Multivariate sign stability across refits and univariate significance are different axes of evidence; the panel complements rather than duplicates the DE ranking (Appendix F), and a candidate list drawn from both axes is still exploratory until tested independently.

## Appendix F. Top 100 differential probes

**Table F1. Top 100 probes by |Welch t| (relapse vs control means).**

| probe       | t      | mean (relapse) | mean (control) |
|-------------|--------|----------------|----------------|
| 209831_x_at | -6.408 | 1375.015       | 1671.832       |
| 204490_s_at | -6.273 | 736.938        | 1071.305       |
| 209835_x_at | -6.209 | 1275.036       | 2067.777       |
| 212014_x_at | -5.933 | 1089.358       | 1799.588       |
| 210916_s_at | -5.515 | 296.571        | 429.647        |
| 214449_s_at | -5.474 | 165.775        | 219.702        |

|             |        |           |           |
|-------------|--------|-----------|-----------|
| 218787_x_at | -5.455 | 95.116    | 152.383   |
| 207551_s_at | -5.323 | 683.691   | 848.409   |
| 209201_x_at | -5.254 | 740.413   | 1162.143  |
| 209138_x_at | -5.25  | 11982.783 | 24169.922 |
| 202216_x_at | -5.22  | 385.642   | 490.731   |
| 204999_s_at | -5.208 | 185.462   | 248.714   |
| 211302_s_at | -5.151 | 151.264   | 231.144   |
| 205926_at   | -5.139 | 380.128   | 531.0     |
| 221986_s_at | -5.137 | 67.774    | 104.488   |
| 215121_x_at | -5.025 | 10387.289 | 23634.254 |
| 212525_s_at | -5.018 | 231.365   | 292.641   |
| 206636_at   | -5.012 | 102.667   | 146.472   |
| 211336_x_at | -5.004 | 142.538   | 229.686   |
| 221224_s_at | -4.964 | 50.107    | 79.448    |
| 214543_x_at | -4.956 | 221.248   | 311.615   |
| 217251_x_at | -4.953 | 32.694    | 48.304    |
| 205457_at   | -4.942 | 152.449   | 202.777   |
| 200956_s_at | -4.936 | 490.941   | 675.085   |
| 209374_s_at | -4.931 | 1733.09   | 4599.721  |
| 205049_s_at | -4.929 | 180.026   | 315.003   |
| 210019_at   | -4.92  | 34.522    | 59.676    |
| 209539_at   | -4.909 | 806.645   | 1078.686  |
| 204489_s_at | -4.878 | 1111.478  | 1588.056  |
| 211528_x_at | -4.871 | 6132.666  | 8976.993  |
| 214677_x_at | -4.861 | 14340.777 | 25025.557 |
| 208624_s_at | -4.837 | 983.076   | 1236.112  |
| 211251_x_at | -4.834 | 407.615   | 509.451   |
| 211430_s_at | -4.83  | 12659.889 | 24251.795 |
| 204197_s_at | -4.826 | 331.874   | 450.995   |
| 221651_x_at | -4.821 | 15523.725 | 26150.623 |
| 208894_at   | -4.814 | 4884.864  | 7495.861  |
| 203388_at   | -4.795 | 234.029   | 339.127   |
| 211919_s_at | -4.793 | 904.016   | 1281.33   |
| 209312_x_at | -4.789 | 9314.111  | 13219.675 |
| 200628_s_at | -4.778 | 352.499   | 593.75    |
| 217436_x_at | -4.767 | 2170.968  | 3130.486  |
| 208885_at   | -4.752 | 632.38    | 924.518   |
| 208594_x_at | -4.743 | 178.857   | 227.539   |
| 207085_x_at | -4.732 | 48.251    | 81.689    |
| 200707_at   | -4.727 | 899.556   | 1230.651  |
| 208791_at   | -4.726 | 1933.209  | 3470.843  |
| 211902_x_at | -4.726 | 235.149   | 336.217   |
| 221477_s_at | -4.711 | 823.1     | 1210.391  |
| 205585_at   | -4.71  | 152.596   | 201.641   |
| 320_at      | -4.71  | 221.39    | 281.025   |
| 211868_x_at | -4.701 | 158.699   | 371.805   |
| 220483_s_at | -4.701 | 128.215   | 217.627   |
| 201039_s_at | -4.688 | 647.596   | 832.677   |
| 201369_s_at | -4.675 | 446.662   | 639.321   |
| 204198_s_at | -4.665 | 388.601   | 612.626   |
| 218232_at   | -4.661 | 724.999   | 1110.725  |
| 211911_x_at | -4.651 | 10693.096 | 15123.041 |
| 210982_s_at | -4.65  | 4534.946  | 6921.172  |

|             |        |           |           |
|-------------|--------|-----------|-----------|
| 216984_x_at | -4.65  | 293.21    | 1256.02   |
| 221671_x_at | -4.641 | 15455.822 | 25740.539 |
| 210754_s_at | -4.633 | 408.819   | 581.463   |
| 208656_s_at | -4.63  | 6912.509  | 8338.373  |
| 215193_x_at | -4.625 | 5628.952  | 8384.064  |
| 221432_s_at | -4.611 | 331.123   | 430.401   |
| 212900_at   | 4.602  | 929.987   | 758.853   |
| 204192_at   | -4.6   | 184.541   | 300.233   |
| 207616_s_at | -4.592 | 1039.98   | 1318.536  |
| 209151_x_at | -4.591 | 25.713    | 45.251    |
| 210785_s_at | -4.588 | 334.671   | 442.19    |
| 207571_x_at | -4.577 | 297.874   | 407.078   |
| 217478_s_at | -4.565 | 1445.798  | 1979.391  |
| 215379_x_at | -4.564 | 4353.471  | 11267.554 |
| 209584_x_at | -4.559 | 273.287   | 351.533   |
| 209846_s_at | -4.554 | 427.215   | 608.389   |
| 201564_s_at | -4.544 | 159.746   | 266.847   |
| 211068_x_at | -4.537 | 1314.228  | 1554.457  |
| 209862_s_at | -4.535 | 253.071   | 307.7     |
| 204562_at   | -4.534 | 246.062   | 337.2     |
| 201514_s_at | -4.523 | 437.452   | 551.533   |
| 203524_s_at | -4.509 | 889.643   | 1070.391  |
| 200637_s_at | -4.491 | 574.455   | 840.722   |
| 208961_s_at | -4.488 | 1007.333  | 1350.91   |
| 208931_s_at | -4.469 | 631.465   | 793.208   |
| 215832_x_at | -4.465 | 310.177   | 406.825   |
| 212063_at   | -4.46  | 3534.702  | 4762.68   |
| 209823_x_at | -4.453 | 631.878   | 1029.007  |
| 220894_x_at | -4.452 | 15.628    | 19.816    |
| 202269_x_at | -4.449 | 909.228   | 1535.755  |
| 214836_x_at | -4.449 | 2910.986  | 6398.214  |
| 202247_s_at | -4.448 | 192.512   | 242.544   |
| 210164_at   | -4.443 | 184.564   | 332.623   |
| 209024_s_at | -4.431 | 1376.527  | 1883.116  |
| 214669_x_at | -4.431 | 2808.797  | 6380.449  |
| 210284_s_at | -4.431 | 410.316   | 528.174   |
| 204670_x_at | -4.427 | 8457.225  | 11352.159 |
| 205322_s_at | -4.424 | 221.974   | 287.605   |
| 209380_s_at | 4.421  | 1660.593  | 1102.959  |
| 207419_s_at | -4.42  | 42.736    | 85.736    |
| 217148_x_at | -4.409 | 909.019   | 2864.25   |

## Appendix G. Protocol decision log

D1. Labels parsed from the series-matrix characteristic 'bone relapses (1=yes, 0=no)'; ambiguous rows would have been dropped (none were).

D2. Feature selection moved INSIDE each CV fold after the first negative result showed split sensitivity; full-dataset selection is leakage; it nevertheless appears in the auxiliary inventories audited on October 1 and 7. Those inventory outputs are not fully fold-internal estimates.

D3. AUC chosen over accuracy after noting 75.9% majority accuracy; accuracy reported only alongside AUC.

D4. Graph arm built on train folds only, because a full-data co-expression graph leaks test correlations into features.

D5. Forty bootstrap refits report empirical sign frequency. Dependence among refits of the same patients prevents interpreting a simple binomial confidence bound as a validated population sign-consistency interval.

D6. The CNN arm's missing AUC is reported as such rather than estimated post-hoc; accuracy alone is kept with the majority baseline adjacent.

D7. Probe mapping uses the official GPL96 annotation (August 2016), not memory; any probe absent from it is reported unmapped.

## **Appendix H. Panel gene biological notes**

CDH13 - cadherin 13; T-cadherin tumor suppressor, frequently silenced by methylation in breast tumors

NPPC - natriuretic peptide C; implicated in angiocrine signaling and reported in metastasis contexts

PASK - PAS-domain serine/threonine kinase; metabolic sensor linking glucose to translation

WFDC1 - WAP four-disulfide core domain 1; secreted protease-inhibitor family, stromal signaling

GLTSCR1 - glioma tumor suppressor candidate region gene 1; chromatin-associated (GBAF complex)

POF1B - premature ovarian failure 1B; actin-binding, candidate biological lead only; novelty has not been independently established

CWF19L1 - cell-cycle control protein homolog; RNA-processing association

EEF2KMT - eEF2 lysine methyltransferase; translational control

COX11 - cytochrome c oxidase copper chaperone; mitochondrial respiration

FUT9 - fucosyltransferase 9; Lewis-x glycan synthesis, glycosylation-driven adhesion changes

CALML3 - calmodulin-like 3; calcium signaling, epithelial differentiation marker

ALDH3A2 - aldehyde dehydrogenase 3A2; lipid aldehyde detoxification

PSMB1 - proteasome beta 1; protein turnover, stress response

CMC4 - C-X9-C motif containing 4; mitochondrial, poorly characterized

BLZF1 - basic leucine zipper nuclear factor 1; transcriptional regulation, Golgi-associated

Notes are orientation, not claims of mechanism; the panel's falsifiable content is the direction stability itself (Table 2).

## Appendix I. The modeling feature pool: top 1,000 probes by |t|

The complete univariate ranking from which every per-fold selection draws (this table is computed on the full cohort for documentation; some modeling selections were fold-internal; global inventory preprocessing remains a caveat).

**Table I1. Top 1,000 probes by |Welch t| with group means.**

| probe       | t      | mean (relapse) | mean (control) |
|-------------|--------|----------------|----------------|
| 209831_x_at | -6.408 | 1375.015       | 1671.832       |
| 204490_s_at | -6.273 | 736.938        | 1071.305       |
| 209835_x_at | -6.209 | 1275.036       | 2067.777       |
| 212014_x_at | -5.933 | 1089.358       | 1799.588       |
| 210916_s_at | -5.515 | 296.571        | 429.647        |
| 214449_s_at | -5.474 | 165.775        | 219.702        |
| 218787_x_at | -5.455 | 95.116         | 152.383        |
| 207551_s_at | -5.323 | 683.691        | 848.409        |
| 209201_x_at | -5.254 | 740.413        | 1162.143       |
| 209138_x_at | -5.25  | 11982.783      | 24169.922      |
| 202216_x_at | -5.22  | 385.642        | 490.731        |
| 204999_s_at | -5.208 | 185.462        | 248.714        |
| 211302_s_at | -5.151 | 151.264        | 231.144        |
| 205926_at   | -5.139 | 380.128        | 531.0          |
| 221986_s_at | -5.137 | 67.774         | 104.488        |
| 215121_x_at | -5.025 | 10387.289      | 23634.254      |
| 212525_s_at | -5.018 | 231.365        | 292.641        |
| 206636_at   | -5.012 | 102.667        | 146.472        |
| 211336_x_at | -5.004 | 142.538        | 229.686        |
| 221224_s_at | -4.964 | 50.107         | 79.448         |
| 214543_x_at | -4.956 | 221.248        | 311.615        |
| 217251_x_at | -4.953 | 32.694         | 48.304         |
| 205457_at   | -4.942 | 152.449        | 202.777        |
| 200956_s_at | -4.936 | 490.941        | 675.085        |
| 209374_s_at | -4.931 | 1733.09        | 4599.721       |
| 205049_s_at | -4.929 | 180.026        | 315.003        |
| 210019_at   | -4.92  | 34.522         | 59.676         |
| 209539_at   | -4.909 | 806.645        | 1078.686       |
| 204489_s_at | -4.878 | 1111.478       | 1588.056       |
| 211528_x_at | -4.871 | 6132.666       | 8976.993       |
| 214677_x_at | -4.861 | 14340.777      | 25025.557      |
| 208624_s_at | -4.837 | 983.076        | 1236.112       |
| 211251_x_at | -4.834 | 407.615        | 509.451        |
| 211430_s_at | -4.83  | 12659.889      | 24251.795      |
| 204197_s_at | -4.826 | 331.874        | 450.995        |
| 221651_x_at | -4.821 | 15523.725      | 26150.623      |
| 208894_at   | -4.814 | 4884.864       | 7495.861       |
| 203388_at   | -4.795 | 234.029        | 339.127        |
| 211919_s_at | -4.793 | 904.016        | 1281.33        |
| 209312_x_at | -4.789 | 9314.111       | 13219.675      |
| 200628_s_at | -4.778 | 352.499        | 593.75         |
| 217436_x_at | -4.767 | 2170.968       | 3130.486       |
| 208885_at   | -4.752 | 632.38         | 924.518        |

|             |        |           |           |
|-------------|--------|-----------|-----------|
| 208594_x_at | -4.743 | 178.857   | 227.539   |
| 207085_x_at | -4.732 | 48.251    | 81.689    |
| 200707_at   | -4.727 | 899.556   | 1230.651  |
| 208791_at   | -4.726 | 1933.209  | 3470.843  |
| 211902_x_at | -4.726 | 235.149   | 336.217   |
| 221477_s_at | -4.711 | 823.1     | 1210.391  |
| 205585_at   | -4.71  | 152.596   | 201.641   |
| 320_at      | -4.71  | 221.39    | 281.025   |
| 211868_x_at | -4.701 | 158.699   | 371.805   |
| 220483_s_at | -4.701 | 128.215   | 217.627   |
| 201039_s_at | -4.688 | 647.596   | 832.677   |
| 201369_s_at | -4.675 | 446.662   | 639.321   |
| 204198_s_at | -4.665 | 388.601   | 612.626   |
| 218232_at   | -4.661 | 724.999   | 1110.725  |
| 211911_x_at | -4.651 | 10693.096 | 15123.041 |
| 210982_s_at | -4.65  | 4534.946  | 6921.172  |
| 216984_x_at | -4.65  | 293.21    | 1256.02   |
| 221671_x_at | -4.641 | 15455.822 | 25740.539 |
| 210754_s_at | -4.633 | 408.819   | 581.463   |
| 208656_s_at | -4.63  | 6912.509  | 8338.373  |
| 215193_x_at | -4.625 | 5628.952  | 8384.064  |
| 221432_s_at | -4.611 | 331.123   | 430.401   |
| 212900_at   | 4.602  | 929.987   | 758.853   |
| 204192_at   | -4.6   | 184.541   | 300.233   |
| 207616_s_at | -4.592 | 1039.98   | 1318.536  |
| 209151_x_at | -4.591 | 25.713    | 45.251    |
| 210785_s_at | -4.588 | 334.671   | 442.19    |
| 207571_x_at | -4.577 | 297.874   | 407.078   |
| 217478_s_at | -4.565 | 1445.798  | 1979.391  |
| 215379_x_at | -4.564 | 4353.471  | 11267.554 |
| 209584_x_at | -4.559 | 273.287   | 351.533   |
| 209846_s_at | -4.554 | 427.215   | 608.389   |
| 201564_s_at | -4.544 | 159.746   | 266.847   |
| 211068_x_at | -4.537 | 1314.228  | 1554.457  |
| 209862_s_at | -4.535 | 253.071   | 307.7     |
| 204562_at   | -4.534 | 246.062   | 337.2     |
| 201514_s_at | -4.523 | 437.452   | 551.533   |
| 203524_s_at | -4.509 | 889.643   | 1070.391  |
| 200637_s_at | -4.491 | 574.455   | 840.722   |
| 208961_s_at | -4.488 | 1007.333  | 1350.91   |
| 208931_s_at | -4.469 | 631.465   | 793.208   |
| 215832_x_at | -4.465 | 310.177   | 406.825   |
| 212063_at   | -4.46  | 3534.702  | 4762.68   |
| 209823_x_at | -4.453 | 631.878   | 1029.007  |
| 220894_x_at | -4.452 | 15.628    | 19.816    |
| 202269_x_at | -4.449 | 909.228   | 1535.755  |
| 214836_x_at | -4.449 | 2910.986  | 6398.214  |
| 202247_s_at | -4.448 | 192.512   | 242.544   |
| 210164_at   | -4.443 | 184.564   | 332.623   |
| 209024_s_at | -4.431 | 1376.527  | 1883.116  |
| 214669_x_at | -4.431 | 2808.797  | 6380.449  |
| 210284_s_at | -4.431 | 410.316   | 528.174   |
| 204670_x_at | -4.427 | 8457.225  | 11352.159 |

|                             |        |           |           |
|-----------------------------|--------|-----------|-----------|
| 205322_s_at                 | -4.424 | 221.974   | 287.605   |
| 209380_s_at                 | 4.421  | 1660.593  | 1102.959  |
| 207419_s_at                 | -4.42  | 42.736    | 85.736    |
| 217148_x_at                 | -4.409 | 909.019   | 2864.25   |
| 204698_at                   | -4.407 | 274.268   | 525.095   |
| 217456_x_at                 | -4.407 | 2851.627  | 3508.919  |
| 218805_at                   | -4.398 | 600.355   | 808.269   |
| 200866_s_at                 | -4.393 | 2597.994  | 3712.696  |
| 201950_x_at                 | -4.389 | 1423.538  | 1735.34   |
| 205014_at                   | -4.385 | 108.42    | 170.547   |
| 217545_at                   | -4.384 | 21.999    | 31.732    |
| AFFX-HSAC07/<br>X00351_M_at | -4.381 | 13259.074 | 17904.639 |
| 211799_x_at                 | -4.371 | 2342.215  | 3632.303  |
| 210869_s_at                 | -4.364 | 526.951   | 692.792   |
| 205320_at                   | -4.363 | 78.499    | 107.957   |
| 203921_at                   | -4.355 | 178.355   | 239.216   |
| 216708_x_at                 | -4.347 | 47.312    | 72.287    |
| 218478_s_at                 | 4.347  | 1033.577  | 845.189   |
| 204116_at                   | -4.337 | 466.739   | 717.871   |
| 32088_at                    | 4.325  | 80.636    | 61.455    |
| 212263_at                   | -4.323 | 595.313   | 764.957   |
| 208729_x_at                 | -4.318 | 8273.566  | 11726.039 |
| 218429_s_at                 | -4.315 | 688.629   | 848.659   |
| 219454_at                   | -4.3   | 256.838   | 387.526   |
| 208306_x_at                 | -4.294 | 11030.797 | 14726.848 |
| 210549_s_at                 | -4.293 | 28.791    | 45.234    |
| 221725_at                   | -4.289 | 1109.077  | 1322.867  |
| 210086_at                   | -4.271 | 29.91     | 45.081    |
| 200706_s_at                 | -4.265 | 1356.064  | 1778.967  |
| 216573_at                   | -4.258 | 46.294    | 88.244    |
| 206833_s_at                 | -4.257 | 400.322   | 485.081   |
| 211004_s_at                 | -4.254 | 253.164   | 305.141   |
| 220939_s_at                 | 4.252  | 1705.383  | 1482.585  |
| 211529_x_at                 | -4.251 | 6623.136  | 9219.072  |
| 202531_at                   | -4.249 | 729.048   | 915.738   |
| 221348_at                   | -4.247 | 34.71     | 53.206    |
| 211797_s_at                 | -4.243 | 93.542    | 147.837   |
| 217138_x_at                 | -4.241 | 18.725    | 45.123    |
| 203332_s_at                 | -4.237 | 650.259   | 821.501   |
| 218085_at                   | 4.236  | 1617.729  | 1273.854  |
| 212671_s_at                 | -4.234 | 1227.161  | 2004.201  |
| 215706_x_at                 | -4.233 | 983.897   | 1273.822  |
| 205462_s_at                 | -4.231 | 89.712    | 134.424   |
| 205778_at                   | -4.23  | 57.612    | 156.482   |
| 208071_s_at                 | -4.226 | 30.761    | 48.831    |
| 211530_x_at                 | -4.226 | 771.268   | 1203.861  |
| 202324_s_at                 | 4.225  | 2326.78   | 1864.908  |
| 219480_at                   | -4.22  | 82.154    | 118.434   |
| 214768_x_at                 | -4.22  | 544.528   | 1819.277  |
| 216365_x_at                 | -4.216 | 59.426    | 226.05    |
| 207893_at                   | -4.213 | 22.429    | 35.175    |
| 214551_s_at                 | -4.208 | 20.375    | 36.893    |



|             |        |           |           |
|-------------|--------|-----------|-----------|
| 205671_s_at | -4.197 | 149.303   | 275.501   |
| 211210_x_at | -4.195 | 137.69    | 191.286   |
| 210214_s_at | -4.194 | 217.661   | 264.733   |
| 33304_at    | -4.191 | 716.352   | 993.359   |
| 211644_x_at | -4.186 | 746.7     | 2403.924  |
| 210972_x_at | -4.178 | 376.925   | 529.399   |
| 205237_at   | -4.177 | 166.068   | 210.257   |
| 205569_at   | -4.176 | 263.777   | 492.574   |
| 214916_x_at | -4.175 | 254.22    | 564.541   |
| 213502_x_at | -4.174 | 2404.917  | 3687.606  |
| 203547_at   | -4.162 | 503.864   | 659.834   |
| 205267_at   | -4.159 | 283.284   | 597.473   |
| 209034_at   | -4.156 | 803.516   | 1064.06   |
| 209203_s_at | -4.151 | 156.82    | 200.319   |
| 201161_s_at | -4.149 | 916.617   | 1262.217  |
| 216511_s_at | -4.145 | 289.538   | 394.275   |
| 211658_at   | -4.142 | 606.064   | 780.875   |
| 215243_s_at | -4.122 | 202.809   | 266.899   |
| 216640_s_at | -4.115 | 2359.805  | 3047.218  |
| 209843_s_at | -4.113 | 50.48     | 90.819    |
| 208793_x_at | -4.113 | 286.074   | 390.656   |
| 220941_s_at | -4.112 | 166.394   | 212.588   |
| 209619_at   | -4.11  | 10662.142 | 13829.485 |
| 220624_s_at | -4.108 | 16.286    | 75.8      |
| 214617_at   | -4.103 | 228.532   | 335.136   |
| 214459_x_at | -4.1   | 22275.117 | 26930.871 |
| 203218_at   | 4.098  | 1183.7    | 809.707   |
| 209968_s_at | -4.095 | 10.723    | 13.5      |
| 217979_at   | 4.095  | 8498.752  | 6323.637  |
| 222230_s_at | 4.092  | 3115.791  | 2620.375  |
| 204552_at   | -4.084 | 187.499   | 231.904   |
| 215633_x_at | -4.076 | 255.377   | 339.934   |
| 212413_at   | -4.074 | 178.951   | 228.052   |
| 217943_s_at | -4.07  | 653.935   | 781.051   |
| 214730_s_at | -4.07  | 1695.002  | 2055.324  |
| 37145_at    | -4.069 | 120.991   | 209.989   |
| 201965_s_at | -4.067 | 579.155   | 680.069   |
| 210285_x_at | -4.066 | 229.752   | 344.911   |
| 222077_s_at | 4.065  | 892.035   | 674.886   |
| 205495_s_at | -4.063 | 155.546   | 237.222   |
| 209406_at   | -4.058 | 155.629   | 210.731   |
| 32723_at    | 4.055  | 222.567   | 187.982   |
| 206125_s_at | -4.05  | 216.348   | 314.893   |
| 207734_at   | -4.047 | 283.942   | 389.036   |
| 213926_s_at | -4.043 | 150.658   | 206.864   |
| 216033_s_at | -4.037 | 243.058   | 340.443   |
| 206359_at   | -4.033 | 81.162    | 118.664   |
| 208433_s_at | -4.033 | 159.984   | 218.67    |
| 201770_at   | -4.032 | 1138.382  | 1336.865  |
| 209725_at   | -4.03  | 388.799   | 448.561   |
| 203470_s_at | -4.028 | 200.872   | 291.065   |
| 207740_s_at | -4.027 | 216.684   | 314.758   |
| 203429_s_at | 4.026  | 1339.33   | 997.222   |

|             |        |          |          |
|-------------|--------|----------|----------|
| 207971_s_at | -4.023 | 200.923  | 240.481  |
| 203157_s_at | -4.019 | 189.396  | 256.871  |
| 208629_s_at | -4.018 | 769.326  | 928.932  |
| 209264_s_at | -4.018 | 253.399  | 342.987  |
| 204132_s_at | -4.017 | 288.016  | 409.0    |
| 221434_s_at | 4.016  | 5434.302 | 4316.178 |
| 211581_x_at | -4.009 | 248.932  | 327.464  |
| 213649_at   | -4.005 | 691.456  | 854.803  |
| 201073_s_at | -4.001 | 277.828  | 356.766  |
| 211211_x_at | -3.997 | 76.893   | 115.823  |
| 209671_x_at | -3.994 | 271.026  | 413.252  |
| 218461_at   | 3.991  | 802.374  | 627.7    |
| 217235_x_at | -3.987 | 393.31   | 936.107  |
| 208792_s_at | -3.985 | 2307.938 | 3685.201 |
| 209026_x_at | -3.982 | 6349.422 | 7991.125 |
| 206150_at   | -3.981 | 268.068  | 420.803  |
| 208960_s_at | -3.981 | 221.575  | 343.598  |
| 216598_s_at | -3.977 | 573.799  | 1050.767 |
| 203530_s_at | -3.975 | 755.486  | 875.84   |
| 212149_at   | 3.975  | 3705.361 | 2912.669 |
| 221825_at   | 3.973  | 1141.604 | 929.196  |
| 201368_at   | -3.968 | 3911.655 | 4856.693 |
| 204060_s_at | -3.968 | 420.039  | 562.954  |
| 211638_at   | -3.964 | 23.588   | 35.805   |
| 205692_s_at | -3.964 | 225.513  | 358.507  |
| 200904_at   | -3.964 | 1278.743 | 1673.116 |
| 200601_at   | -3.957 | 2543.439 | 3274.634 |
| 202867_s_at | -3.955 | 607.91   | 702.557  |
| 203183_s_at | -3.955 | 262.277  | 305.371  |
| 206220_s_at | -3.952 | 34.142   | 51.568   |
| 200726_at   | 3.949  | 3309.946 | 2909.058 |
| 179_at      | -3.945 | 683.167  | 786.467  |
| 202902_s_at | -3.938 | 657.597  | 936.721  |
| 203176_s_at | -3.934 | 379.407  | 447.487  |
| 217960_s_at | -3.932 | 257.68   | 338.732  |
| 210667_s_at | -3.931 | 319.583  | 403.093  |
| 212528_at   | -3.931 | 370.088  | 432.363  |
| 211949_s_at | -3.931 | 361.743  | 418.452  |
| 216207_x_at | -3.929 | 1691.691 | 4138.462 |
| 201656_at   | -3.928 | 509.904  | 732.221  |
| 217442_at   | -3.928 | 13.468   | 16.365   |
| 209764_at   | -3.927 | 75.03    | 112.204  |
| 211641_x_at | -3.926 | 180.806  | 345.446  |
| 212370_x_at | -3.926 | 1504.625 | 1718.394 |
| 220387_s_at | -3.926 | 141.69   | 194.115  |
| 205205_at   | -3.925 | 256.506  | 312.142  |
| 221430_s_at | -3.925 | 308.736  | 363.821  |
| 204821_at   | -3.925 | 223.078  | 289.353  |
| 201708_s_at | -3.923 | 370.839  | 463.672  |
| 211645_x_at | -3.922 | 918.432  | 2843.412 |
| 201565_s_at | -3.921 | 1730.075 | 2091.258 |
| 208598_s_at | -3.919 | 1889.617 | 2182.711 |
| 215946_x_at | -3.918 | 2443.4   | 5753.307 |

|                             |        |           |           |
|-----------------------------|--------|-----------|-----------|
| 201134_x_at                 | 3.911  | 14073.179 | 11838.088 |
| 215176_x_at                 | -3.906 | 1553.254  | 3994.455  |
| 214590_s_at                 | -3.906 | 58.528    | 77.066    |
| 208930_s_at                 | -3.905 | 396.945   | 528.386   |
| 203331_s_at                 | -3.903 | 13.064    | 17.09     |
| 210347_s_at                 | -3.898 | 280.752   | 409.806   |
| 220757_s_at                 | -3.896 | 398.004   | 469.088   |
| 218105_s_at                 | -3.895 | 451.144   | 594.804   |
| 213352_at                   | -3.894 | 236.587   | 277.523   |
| 210692_s_at                 | -3.893 | 113.429   | 166.943   |
| 204269_at                   | -3.891 | 369.016   | 481.835   |
| 202935_s_at                 | -3.886 | 490.538   | 691.525   |
| 218947_s_at                 | -3.886 | 491.636   | 574.08    |
| 212474_at                   | 3.884  | 1135.794  | 950.197   |
| 212886_at                   | -3.88  | 171.645   | 232.535   |
| 218909_at                   | 3.876  | 809.339   | 668.679   |
| 222062_at                   | -3.873 | 236.22    | 377.49    |
| 205452_at                   | 3.869  | 749.922   | 648.323   |
| 213254_at                   | -3.864 | 388.888   | 461.723   |
| 201421_s_at                 | -3.863 | 762.935   | 907.591   |
| 203743_s_at                 | 3.861  | 1288.29   | 1038.726  |
| 204806_x_at                 | -3.858 | 4592.628  | 6080.033  |
| 200869_at                   | -3.856 | 14395.83  | 17976.283 |
| 216037_x_at                 | -3.856 | 601.207   | 720.518   |
| AFFX-HSAC07/<br>X00351_5_at | -3.856 | 3920.71   | 6147.176  |
| 211451_s_at                 | -3.856 | 28.536    | 42.98     |
| 209571_at                   | -3.855 | 266.933   | 306.227   |
| 211991_s_at                 | -3.853 | 3513.92   | 4749.168  |
| 210579_s_at                 | -3.852 | 105.355   | 131.207   |
| 215223_s_at                 | -3.845 | 589.894   | 1047.329  |
| 217179_x_at                 | -3.843 | 409.325   | 850.681   |
| 208983_s_at                 | -3.842 | 212.896   | 310.303   |
| 210770_s_at                 | -3.841 | 56.467    | 86.252    |
| 206844_at                   | -3.84  | 7.916     | 9.295     |
| 212617_at                   | 3.837  | 292.467   | 248.757   |
| 203499_at                   | -3.834 | 247.507   | 298.348   |
| 218136_s_at                 | -3.832 | 175.897   | 226.49    |
| 217161_x_at                 | -3.828 | 223.126   | 313.771   |
| 215177_s_at                 | -3.828 | 148.035   | 270.023   |
| 205943_at                   | -3.827 | 236.459   | 417.946   |
| 203692_s_at                 | -3.822 | 291.655   | 368.877   |
| 204341_at                   | -3.821 | 397.068   | 480.935   |
| 217825_s_at                 | -3.821 | 539.246   | 629.35    |
| 201769_at                   | 3.819  | 1435.423  | 1228.315  |
| 211546_x_at                 | -3.819 | 149.151   | 180.125   |
| 213197_at                   | -3.816 | 37.151    | 53.356    |
| 202543_s_at                 | 3.816  | 746.648   | 552.728   |
| 219972_s_at                 | 3.815  | 253.448   | 209.45    |
| 217826_s_at                 | -3.814 | 440.23    | 561.985   |
| 216226_at                   | -3.811 | 188.738   | 229.292   |
| 210629_x_at                 | -3.807 | 369.059   | 451.565   |
| 212219_at                   | -3.806 | 1074.88   | 1331.957  |

|                                     |        |           |           |
|-------------------------------------|--------|-----------|-----------|
| 211584_s_at                         | -3.805 | 137.968   | 183.126   |
| 200917_s_at                         | -3.804 | 141.412   | 214.025   |
| 217223_s_at                         | -3.803 | 202.133   | 261.172   |
| 214946_x_at                         | -3.803 | 1121.467  | 1320.435  |
| 200695_at                           | -3.802 | 1007.545  | 1257.59   |
| 217523_at                           | -3.793 | 361.039   | 527.386   |
| 212025_s_at                         | -3.79  | 333.906   | 432.956   |
| 216557_x_at                         | -3.79  | 203.083   | 368.921   |
| 212108_at                           | 3.789  | 1277.662  | 1099.432  |
| 219319_at                           | -3.787 | 37.71     | 49.297    |
| 218842_at                           | 3.784  | 497.284   | 423.183   |
| 211610_at                           | -3.782 | 187.613   | 231.488   |
| 222003_s_at                         | -3.779 | 60.87     | 86.688    |
| 205945_at                           | -3.778 | 149.451   | 198.63    |
| AFFX-<br>HUMISGF3A/<br>M97935_MA_at | -3.778 | 309.687   | 499.714   |
| 204998_s_at                         | -3.776 | 414.345   | 486.946   |
| 200937_s_at                         | -3.772 | 10451.861 | 12117.686 |
| 211286_x_at                         | -3.77  | 93.48     | 127.409   |
| 202150_s_at                         | -3.769 | 212.503   | 266.95    |
| 210015_s_at                         | -3.769 | 105.593   | 136.848   |
| 208627_s_at                         | -3.766 | 3767.158  | 5034.159  |
| 210764_s_at                         | -3.763 | 914.852   | 1297.789  |
| 207741_x_at                         | -3.762 | 332.687   | 557.065   |
| 215338_s_at                         | -3.76  | 299.354   | 373.611   |
| 209890_at                           | -3.756 | 332.33    | 471.521   |
| 202643_s_at                         | -3.755 | 318.452   | 383.328   |
| 206486_at                           | -3.754 | 73.701    | 117.961   |
| 201178_at                           | -3.754 | 2222.052  | 2515.354  |
| 219498_s_at                         | -3.751 | 127.857   | 196.597   |
| 208392_x_at                         | -3.75  | 173.013   | 244.876   |
| 211917_s_at                         | -3.749 | 114.925   | 151.194   |
| 201488_x_at                         | -3.747 | 1686.222  | 1860.317  |
| 208631_s_at                         | -3.741 | 816.283   | 1005.913  |
| 222084_s_at                         | -3.739 | 8.045     | 12.345    |
| 221955_at                           | -3.739 | 6.129     | 7.415     |
| 205812_s_at                         | 3.738  | 7269.014  | 5848.605  |
| 201858_s_at                         | -3.737 | 1094.38   | 1553.363  |
| 211255_x_at                         | -3.735 | 149.771   | 201.8     |
| 200014_s_at                         | 3.734  | 3815.013  | 3321.009  |
| 211488_s_at                         | -3.733 | 79.345    | 102.679   |
| 221658_s_at                         | -3.732 | 91.291    | 135.369   |
| 203732_at                           | 3.73   | 1008.823  | 913.581   |
| 200951_s_at                         | -3.728 | 59.619    | 92.819    |
| 214777_at                           | -3.726 | 332.855   | 1190.42   |
| 214298_x_at                         | -3.725 | 306.568   | 375.924   |
| 202368_s_at                         | -3.724 | 482.187   | 552.87    |
| 203206_at                           | -3.723 | 802.974   | 929.584   |
| 201075_s_at                         | -3.72  | 400.397   | 520.652   |
| 203175_at                           | -3.718 | 1132.952  | 1291.612  |
| 211649_x_at                         | -3.718 | 81.661    | 152.604   |
| 210797_s_at                         | -3.717 | 377.825   | 492.134   |

|                                     |        |          |          |
|-------------------------------------|--------|----------|----------|
| 208032_s_at                         | -3.717 | 14.858   | 27.096   |
| 209078_s_at                         | -3.717 | 578.532  | 711.303  |
| 215431_at                           | -3.709 | 17.729   | 23.919   |
| 219907_at                           | -3.708 | 249.625  | 281.454  |
| 210506_at                           | -3.708 | 18.187   | 26.564   |
| 213550_s_at                         | 3.707  | 391.441  | 296.247  |
| 208829_at                           | -3.705 | 2584.091 | 3228.375 |
| 215714_s_at                         | -3.703 | 1094.362 | 1317.767 |
| 206644_at                           | -3.702 | 64.591   | 84.82    |
| 211634_x_at                         | -3.702 | 157.059  | 365.486  |
| 211582_x_at                         | -3.695 | 341.174  | 438.99   |
| 214581_x_at                         | -3.695 | 109.967  | 163.598  |
| 214941_s_at                         | -3.695 | 474.752  | 618.529  |
| 204933_s_at                         | -3.695 | 30.125   | 93.79    |
| 215949_x_at                         | -3.695 | 72.051   | 182.691  |
| 211536_x_at                         | -3.694 | 318.71   | 388.803  |
| 210514_x_at                         | -3.693 | 2441.345 | 3530.642 |
| 204480_s_at                         | -3.689 | 944.791  | 1150.714 |
| 200008_s_at                         | -3.687 | 867.728  | 1207.643 |
| 215945_s_at                         | -3.684 | 280.413  | 346.255  |
| 206980_s_at                         | -3.683 | 28.175   | 40.735   |
| 200905_x_at                         | -3.681 | 7925.635 | 9928.193 |
| 213650_at                           | -3.681 | 479.997  | 609.802  |
| 208018_s_at                         | -3.68  | 357.736  | 469.562  |
| 202307_s_at                         | -3.679 | 1206.0   | 1832.708 |
| 205409_at                           | -3.678 | 93.575   | 129.883  |
| 203274_at                           | -3.678 | 1416.226 | 1641.86  |
| 211733_x_at                         | 3.677  | 2296.868 | 1964.561 |
| 202596_at                           | 3.675  | 9383.35  | 7600.447 |
| 207795_s_at                         | -3.673 | 141.943  | 183.199  |
| 202082_s_at                         | -3.673 | 252.8    | 300.507  |
| 203492_x_at                         | -3.673 | 330.496  | 390.641  |
| AFFX-<br>HUMISGF3A/<br>M97935_MB_at | -3.672 | 317.757  | 456.105  |
| 204693_at                           | -3.672 | 229.732  | 303.609  |
| 211707_s_at                         | -3.671 | 543.901  | 611.444  |
| 211178_s_at                         | -3.668 | 133.291  | 173.945  |
| 215214_at                           | -3.667 | 193.914  | 691.577  |
| 213490_s_at                         | -3.666 | 112.612  | 159.859  |
| 216252_x_at                         | -3.666 | 156.206  | 202.832  |
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| 216743_at                           | -3.664 | 28.816   | 41.912   |
| 209995_s_at                         | -3.663 | 279.219  | 394.302  |
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| 221286_s_at                         | -3.657 | 273.116  | 596.369  |
| 216602_s_at                         | -3.657 | 218.252  | 300.884  |
| 217379_at                           | -3.654 | 1816.896 | 2291.425 |
| 209666_s_at                         | 3.65   | 329.381  | 284.358  |
| 219193_at                           | -3.649 | 616.667  | 689.205  |
| 217480_x_at                         | -3.648 | 797.539  | 1538.576 |
| 217897_at                           | -3.648 | 465.38   | 586.397  |
| 211643_x_at                         | -3.647 | 682.187  | 1426.209 |

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| 209282_at   | -3.647 | 478.101  | 563.279  |
| 210563_x_at | -3.643 | 569.129  | 707.007  |
| 208843_s_at | 3.642  | 2119.688 | 1804.227 |
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| 210859_x_at | -3.64  | 823.188  | 1033.454 |
| 211139_s_at | -3.64  | 282.593  | 322.647  |
| 202630_at   | 3.638  | 912.722  | 630.338  |
| 212125_at   | -3.637 | 208.22   | 278.305  |
| 209344_at   | -3.637 | 788.264  | 1057.244 |
| 216474_x_at | -3.637 | 373.578  | 606.853  |
| 215382_x_at | -3.635 | 230.791  | 427.171  |
| 205861_at   | -3.633 | 179.643  | 299.971  |
| 210396_s_at | 3.631  | 1620.977 | 1320.654 |
| 221866_at   | -3.63  | 201.287  | 253.469  |
| 210354_at   | -3.628 | 129.428  | 158.592  |
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| 221742_at   | -3.618 | 753.646  | 869.494  |
| 212211_at   | -3.617 | 641.152  | 757.473  |
| 207485_x_at | -3.614 | 92.123   | 124.295  |
| 216401_x_at | -3.612 | 413.887  | 1349.234 |
| 212974_at   | -3.612 | 152.61   | 183.987  |
| 221078_s_at | -3.609 | 76.588   | 101.55   |
| 211537_x_at | -3.609 | 309.08   | 390.332  |
| 210386_s_at | 3.608  | 2151.758 | 1800.892 |
| 211947_s_at | -3.606 | 559.725  | 759.95   |
| 218942_at   | 3.606  | 1182.249 | 993.601  |
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| 217996_at   | -3.598 | 362.188  | 538.445  |
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| 219053_s_at | 3.593  | 1670.271 | 1340.235 |
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| 206649_s_at | -3.588 | 261.839  | 306.949  |
| 50314_i_at  | -3.587 | 212.874  | 273.666  |
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| 206701_x_at | -3.576 | 139.814  | 173.044  |
| 219758_at   | 3.576  | 152.578  | 115.461  |
| 216329_at   | -3.575 | 18.554   | 28.31    |
| 211358_s_at | -3.571 | 298.533  | 353.842  |

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|-------------|--------|-----------|-----------|
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| 206932_at   | -3.56  | 202.216   | 258.198   |
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| 217828_at   | 3.555  | 1306.39   | 1126.115  |
| 36554_at    | -3.554 | 370.087   | 461.595   |
| 210177_at   | -3.552 | 21.658    | 28.059    |
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| 203253_s_at | 3.548  | 778.57    | 621.621   |
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| 201358_s_at | 3.532  | 6423.346  | 5488.681  |
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| 221985_at   | -3.531 | 376.59    | 452.495   |
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| 210786_s_at | -3.525 | 107.154   | 132.514   |
| 217326_x_at | -3.521 | 34.307    | 52.192    |
| 211469_s_at | -3.52  | 31.633    | 48.295    |
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| 220382_s_at | -3.509 | 51.77     | 71.724    |
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| 206133_at   | -3.506 | 435.69    | 595.519   |
| 212379_at   | -3.504 | 231.359   | 305.411   |
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| 200824_at   | -3.5   | 1317.168  | 1859.278  |
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| 203052_at   | -3.476 | 495.416  | 575.214  |
| 202132_at   | -3.47  | 383.925  | 514.91   |
| 202134_s_at | -3.469 | 146.483  | 233.671  |
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| 201575_at   | 3.454  | 1748.79  | 1491.997 |
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| 91816_f_at  | -3.448 | 222.749  | 279.401  |
| 222263_at   | -3.446 | 296.536  | 367.38   |
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| 201201_at   | -3.444 | 4954.885 | 6075.913 |
| 204725_s_at | -3.443 | 532.3    | 654.133  |
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| 217471_at   | -3.441 | 114.459  | 136.545  |
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|-------------|--------|----------|-----------|
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| 218379_at   | -3.436 | 615.003  | 728.345   |
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| 205707_at   | -3.423 | 462.977  | 530.206   |
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| 214152_at   | 3.42   | 504.984  | 420.84    |
| 202058_s_at | -3.42  | 290.238  | 341.845   |
| 210543_s_at | -3.42  | 463.174  | 627.277   |
| 204265_s_at | -3.417 | 521.57   | 643.985   |
| 206226_at   | -3.416 | 107.551  | 133.297   |
| 203714_s_at | 3.415  | 1088.761 | 883.805   |
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| 204545_at   | -3.413 | 94.232   | 152.339   |
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| 221688_s_at | -3.409 | 1056.277 | 1251.868  |
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| 206722_s_at | -3.405 | 190.825  | 243.715   |
| 219971_at   | -3.405 | 43.609   | 64.703    |
| 210849_s_at | 3.404  | 1208.671 | 1095.59   |
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| 211714_x_at | -3.403 | 8641.627 | 10254.078 |
| 213286_at   | -3.403 | 79.275   | 109.991   |
| 220743_at   | -3.403 | 126.968  | 164.083   |
| 209524_at   | 3.402  | 938.826  | 702.041   |
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| 221159_at   | -3.401 | 114.059  | 140.388   |
| 218241_at   | 3.401  | 1509.267 | 1269.744  |
| 208986_at   | 3.4    | 2083.543 | 1792.952  |
| 206203_at   | -3.399 | 51.499   | 70.526    |
| 203104_at   | -3.399 | 519.635  | 646.067   |
| 203751_x_at | -3.395 | 160.801  | 240.489   |
| 214881_s_at | -3.393 | 120.164  | 160.832   |
| 221875_x_at | -3.392 | 6994.911 | 8790.784  |
| 220712_at   | -3.391 | 299.812  | 338.571   |
| 221969_at   | -3.389 | 176.465  | 248.791   |
| 209944_at   | 3.389  | 3242.342 | 2752.73   |
| 221706_s_at | -3.389 | 256.258  | 303.813   |
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| 210029_at   | -3.388 | 412.452  | 728.486   |
| 208074_s_at | -3.386 | 864.313  | 1072.414  |
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| 211136_s_at | -3.385 | 339.964  | 454.754   |
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| 204683_at   | -3.384 | 326.28    | 397.229   |
| 205807_s_at | 3.384  | 1679.842  | 1385.228  |
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| 205063_at   | 3.38   | 222.41    | 179.636   |
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| 211979_at   | 3.378  | 1011.191  | 858.433   |
| 205660_at   | -3.378 | 449.996   | 573.332   |
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| 211796_s_at | -3.377 | 509.994   | 894.986   |
| 204702_s_at | -3.376 | 231.171   | 314.005   |
| 205687_at   | 3.376  | 387.803   | 333.269   |
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| 213957_s_at | -3.372 | 391.741   | 452.815   |
| 206654_s_at | -3.369 | 51.768    | 66.035    |
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| 205683_x_at | -3.367 | 524.686   | 817.606   |
| 201869_s_at | -3.366 | 100.846   | 155.724   |
| 210325_at   | -3.366 | 267.38    | 322.905   |
| 201495_x_at | -3.363 | 9.449     | 11.53     |
| 220603_s_at | -3.363 | 161.275   | 197.29    |
| 205798_at   | -3.362 | 509.059   | 723.868   |
| 205193_at   | -3.361 | 326.965   | 402.25    |
| 202172_at   | 3.361  | 1411.025  | 1193.757  |
| 202001_s_at | 3.359  | 3327.839  | 2752.458  |
| 221293_s_at | -3.357 | 102.201   | 140.307   |
| 210649_s_at | -3.356 | 492.439   | 602.896   |
| 208268_at   | -3.355 | 90.2      | 117.044   |
| 216430_x_at | -3.353 | 36.354    | 104.8     |
| 208087_s_at | -3.352 | 58.255    | 82.729    |
| 219716_at   | -3.352 | 276.57    | 362.699   |
| 205297_s_at | -3.352 | 106.371   | 156.795   |
| 210645_s_at | 3.352  | 4728.647  | 3961.44   |
| 221257_x_at | 3.351  | 822.865   | 706.186   |
| 205485_at   | -3.351 | 57.361    | 118.725   |
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| 221791_s_at | 3.349  | 15235.583 | 12993.454 |
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| 202524_s_at | -3.349 | 514.38    | 737.734   |
| 202879_s_at | -3.347 | 68.241    | 89.819    |
| 209275_s_at | -3.346 | 731.664   | 883.559   |
| 205743_at   | -3.345 | 337.407   | 417.31    |
| 219247_s_at | -3.344 | 123.385   | 196.508   |
| 202214_s_at | -3.342 | 1148.913  | 1305.694  |
| 208854_s_at | -3.342 | 958.875   | 1127.699  |
| 216560_x_at | -3.341 | 113.959   | 551.724   |
| 221795_at   | -3.337 | 139.994   | 233.851   |
| 201642_at   | -3.337 | 3066.342  | 3465.236  |
| 203059_s_at | -3.335 | 254.691   | 322.529   |
| 216632_at   | -3.335 | 9.306     | 12.502    |
| 207515_s_at | -3.332 | 472.93    | 527.231   |
| 211676_s_at | -3.332 | 926.307   | 1132.432  |
| 202957_at   | -3.331 | 840.172   | 1069.415  |

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| 202624_s_at | -3.331 | 525.844  | 639.337  |
| 208716_s_at | 3.33   | 4185.507 | 3540.598 |
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| 219235_s_at | -3.329 | 452.928  | 535.047  |
| 211039_at   | -3.329 | 33.48    | 48.157   |
| 210756_s_at | -3.329 | 336.343  | 471.972  |
| 212552_at   | -3.329 | 1180.188 | 1400.311 |
| 218830_at   | 3.328  | 2172.711 | 1898.454 |
| 202715_at   | -3.326 | 408.967  | 495.182  |
| 216262_s_at | -3.325 | 424.054  | 504.661  |
| 207069_s_at | 3.322  | 244.752  | 181.688  |
| 219717_at   | -3.321 | 370.699  | 452.706  |
| 201529_s_at | -3.32  | 776.988  | 975.997  |
| 210677_at   | -3.319 | 37.772   | 53.995   |
| 217898_at   | 3.319  | 2299.028 | 2057.588 |
| 202206_at   | -3.318 | 312.049  | 401.956  |
| 206404_at   | -3.318 | 95.713   | 143.532  |
| 217405_x_at | -3.317 | 130.341  | 156.086  |
| 218252_at   | 3.317  | 722.683  | 597.73   |
| 210011_s_at | -3.317 | 859.971  | 979.86   |
| 214865_at   | -3.316 | 9.746    | 11.752   |
| 209407_s_at | -3.313 | 405.62   | 507.683  |
| 214338_at   | -3.313 | 221.862  | 262.063  |
| 208902_s_at | -3.312 | 75.284   | 94.855   |
| 209761_s_at | -3.311 | 658.594  | 883.869  |
| 212748_at   | -3.311 | 287.438  | 327.316  |
| 203061_s_at | -3.31  | 35.626   | 59.005   |
| 51192_at    | 3.31   | 1080.429 | 807.15   |
| 219773_at   | 3.309  | 447.503  | 346.108  |
| 217227_x_at | -3.308 | 263.271  | 596.317  |
| 204449_at   | 3.307  | 235.293  | 201.074  |
| 204990_s_at | -3.306 | 509.017  | 699.606  |
| 202824_s_at | 3.306  | 2800.974 | 2221.56  |
| 210038_at   | -3.306 | 240.916  | 301.357  |
| 210030_at   | -3.303 | 37.575   | 54.649   |
| 207505_at   | -3.303 | 128.885  | 158.814  |
| 216269_s_at | -3.301 | 131.081  | 162.596  |
| 221517_s_at | -3.301 | 522.501  | 611.565  |
| 207941_s_at | 3.3    | 4106.808 | 3620.488 |
| 205822_s_at | -3.298 | 126.541  | 157.599  |
| 212171_x_at | -3.298 | 1204.037 | 1534.196 |
| 220684_at   | -3.298 | 193.071  | 223.545  |
| 216892_at   | -3.297 | 79.52    | 133.866  |
| 39318_at    | -3.297 | 164.812  | 238.668  |
| 218215_s_at | -3.297 | 215.239  | 255.92   |
| 201585_s_at | -3.297 | 757.857  | 857.8    |
| 207036_x_at | -3.297 | 19.884   | 28.751   |
| 217860_at   | -3.296 | 1717.342 | 2008.355 |
| 218668_s_at | 3.293  | 684.822  | 526.975  |
| 204312_x_at | -3.293 | 218.101  | 275.533  |
| 200007_at   | 3.293  | 9103.357 | 8439.176 |
| 214853_s_at | 3.292  | 4333.045 | 3642.548 |
| 214369_s_at | -3.292 | 165.613  | 199.63   |

|             |        |           |           |
|-------------|--------|-----------|-----------|
| 202253_s_at | -3.292 | 203.938   | 256.671   |
| 217281_x_at | -3.289 | 83.532    | 246.286   |
| 201487_at   | -3.289 | 1789.322  | 2424.955  |
| 206326_at   | 3.288  | 527.004   | 370.19    |
| 200672_x_at | -3.286 | 294.703   | 376.732   |
| 208541_x_at | -3.284 | 237.265   | 281.826   |
| 218356_at   | 3.284  | 1016.371  | 867.837   |
| 208938_at   | -3.282 | 281.148   | 341.151   |
| 205788_s_at | 3.281  | 6097.922  | 5338.331  |
| 207777_s_at | -3.28  | 117.361   | 159.327   |
| 221693_s_at | -3.28  | 342.448   | 404.642   |
| 201299_s_at | -3.279 | 183.925   | 235.463   |
| 209211_at   | -3.278 | 103.0     | 145.342   |
| 209152_s_at | -3.276 | 264.303   | 313.8     |
| 204641_at   | 3.276  | 742.067   | 551.726   |
| 200001_at   | -3.272 | 2765.989  | 3393.647  |
| 213761_at   | 3.272  | 377.023   | 314.825   |
| 203093_s_at | -3.271 | 245.249   | 286.757   |
| 211941_s_at | -3.27  | 4136.865  | 4811.56   |
| 210719_s_at | -3.269 | 972.896   | 1211.066  |
| 219309_at   | -3.268 | 71.613    | 93.746    |
| 203662_s_at | -3.265 | 33.801    | 58.739    |
| 206074_s_at | -3.264 | 828.91    | 1059.091  |
| 210915_x_at | -3.264 | 823.128   | 1303.974  |
| 221512_at   | -3.263 | 615.891   | 709.194   |
| 201085_s_at | -3.263 | 1195.994  | 1433.237  |
| 202424_at   | -3.261 | 1321.452  | 1569.206  |
| 201050_at   | -3.261 | 388.175   | 504.415   |
| 208812_x_at | -3.26  | 24635.799 | 27986.666 |
| 205490_x_at | -3.26  | 103.283   | 168.038   |
| 214005_at   | 3.259  | 570.346   | 471.761   |
| 218767_at   | 3.257  | 447.881   | 379.764   |
| 202734_at   | -3.254 | 471.107   | 530.241   |
| 215141_at   | -3.254 | 107.404   | 133.341   |
| 202793_at   | -3.254 | 153.497   | 199.566   |
| 215320_at   | -3.254 | 4.794     | 7.751     |
| 216375_s_at | -3.254 | 107.2     | 137.756   |
| 207610_s_at | -3.253 | 116.606   | 145.733   |
| 214820_at   | 3.252  | 271.877   | 219.981   |
| 203345_s_at | -3.252 | 689.71    | 780.949   |
| 217894_at   | 3.251  | 787.244   | 576.828   |
| 202748_at   | -3.251 | 1137.985  | 1417.489  |
| 204733_at   | -3.25  | 233.868   | 450.854   |
| 215777_at   | -3.25  | 76.593    | 150.994   |
| 210809_s_at | 3.249  | 17241.197 | 14237.523 |
| 202692_s_at | -3.248 | 417.099   | 470.255   |
| 219471_at   | -3.246 | 237.496   | 306.363   |
| 215171_s_at | 3.245  | 3404.517  | 2855.968  |
| 206574_s_at | -3.245 | 158.014   | 227.387   |
| 221821_s_at | 3.245  | 881.413   | 794.029   |
| 212416_at   | 3.243  | 3004.296  | 2595.246  |
| 218534_s_at | 3.239  | 652.385   | 569.993   |
| 218582_at   | -3.238 | 683.142   | 769.079   |

|             |        |          |           |
|-------------|--------|----------|-----------|
| 211403_x_at | -3.238 | 95.522   | 119.952   |
| 211913_s_at | -3.237 | 158.001  | 184.609   |
| 209196_at   | -3.237 | 212.43   | 257.223   |
| 211269_s_at | -3.237 | 180.796  | 217.738   |
| 202534_x_at | 3.237  | 871.835  | 721.495   |
| 217099_s_at | -3.237 | 241.484  | 286.506   |
| 41329_at    | 3.237  | 703.371  | 563.154   |
| 204454_at   | -3.235 | 1239.084 | 1563.772  |
| 200876_s_at | -3.234 | 3506.284 | 3919.109  |
| 201994_at   | 3.233  | 13031.08 | 11422.045 |
| 219256_s_at | -3.233 | 247.441  | 289.379   |
| 213293_s_at | -3.233 | 1077.852 | 1326.574  |
| 210084_x_at | -3.232 | 232.372  | 416.442   |
| 215078_at   | -3.23  | 50.806   | 79.556    |
| 221849_s_at | -3.228 | 193.4    | 230.279   |
| 202663_at   | -3.228 | 103.903  | 129.835   |
| 214277_at   | 3.227  | 60.593   | 43.507    |
| 209878_s_at | -3.225 | 147.923  | 189.715   |
| 203050_at   | 3.224  | 1170.316 | 990.917   |
| 204563_at   | -3.224 | 293.206  | 451.938   |
| 221698_s_at | -3.224 | 416.485  | 547.882   |
| 219753_at   | -3.223 | 130.426  | 161.736   |
| 201093_x_at | -3.222 | 1515.31  | 1744.511  |
| 219014_at   | -3.221 | 380.141  | 550.108   |
| 213424_at   | 3.221  | 160.436  | 130.56    |
| 218600_at   | -3.219 | 132.449  | 172.555   |
| 209685_s_at | -3.218 | 321.499  | 417.849   |
| 221242_at   | -3.217 | 74.514   | 96.118    |
| 218989_x_at | 3.217  | 577.559  | 500.523   |
| 204662_at   | 3.216  | 720.825  | 590.331   |
| 210133_at   | -3.216 | 121.616  | 153.224   |
| 219452_at   | -3.216 | 17.183   | 33.853    |
| 218003_s_at | 3.215  | 2007.681 | 1715.525  |
| 212337_at   | -3.214 | 795.848  | 1019.67   |
| 202175_at   | -3.213 | 679.365  | 779.025   |
| 204496_at   | 3.213  | 444.128  | 371.078   |
| 209797_at   | 3.212  | 868.839  | 737.751   |
| 214439_x_at | -3.211 | 465.951  | 577.985   |
| 32042_at    | -3.211 | 146.733  | 170.341   |
| 201420_s_at | -3.21  | 412.397  | 489.587   |
| 216510_x_at | -3.21  | 108.175  | 320.608   |
| 217729_s_at | -3.207 | 635.339  | 774.242   |
| 201234_at   | -3.205 | 1102.894 | 1275.183  |
| 210959_s_at | -3.205 | 152.677  | 200.324   |
| 220425_x_at | -3.205 | 213.148  | 363.059   |
| 206118_at   | -3.204 | 308.606  | 370.159   |
| 208991_at   | -3.204 | 3248.904 | 3856.903  |
| 204724_s_at | -3.202 | 114.62   | 323.214   |
| 203508_at   | -3.201 | 341.62   | 428.843   |
| 207841_at   | -3.201 | 101.464  | 126.058   |
| 221532_s_at | 3.201  | 2224.41  | 1960.48   |
| 209670_at   | -3.198 | 563.848  | 825.977   |
| 204279_at   | -3.198 | 1213.543 | 1638.136  |

|             |        |          |          |
|-------------|--------|----------|----------|
| 216177_at   | -3.198 | 222.741  | 272.193  |
| 214830_at   | 3.197  | 377.022  | 316.687  |
| 217297_s_at | -3.196 | 151.433  | 191.14   |
| 221291_at   | -3.196 | 26.109   | 38.65    |
| 209681_at   | 3.195  | 1629.21  | 1140.93  |
| 204078_at   | 3.194  | 418.009  | 300.913  |
| 207996_s_at | 3.194  | 720.464  | 486.208  |
| 215346_at   | -3.193 | 192.626  | 242.488  |
| 207736_s_at | -3.192 | 6.238    | 8.644    |
| 35150_at    | -3.19  | 591.648  | 671.829  |
| 212613_at   | -3.189 | 195.567  | 264.941  |
| 203721_s_at | 3.188  | 2048.442 | 1571.451 |
| 204309_at   | -3.188 | 74.264   | 92.394   |
| 211771_s_at | -3.187 | 29.203   | 42.488   |
| 218985_at   | 3.187  | 199.2    | 153.871  |
| 201064_s_at | -3.187 | 1721.674 | 2008.152 |
| 222175_s_at | -3.185 | 468.356  | 552.117  |
| 203565_s_at | 3.184  | 430.858  | 341.932  |
| 202805_s_at | -3.184 | 219.652  | 272.376  |
| 211876_x_at | -3.184 | 9.325    | 13.871   |
| 201513_at   | 3.184  | 434.878  | 385.068  |
| 218262_at   | 3.183  | 911.21   | 767.236  |
| 204320_at   | 3.182  | 3880.156 | 2387.76  |
| 212078_s_at | -3.182 | 371.02   | 432.845  |
| 214265_at   | -3.182 | 131.491  | 157.321  |
| 213203_at   | 3.181  | 1267.101 | 1117.243 |
| 221483_s_at | 3.18   | 3710.921 | 3314.237 |
| 209626_s_at | -3.179 | 207.425  | 257.896  |
| 221529_s_at | -3.178 | 399.99   | 491.721  |
| 202271_at   | 3.178  | 1329.258 | 1118.729 |
| 207457_s_at | -3.176 | 113.568  | 139.937  |
| 219997_s_at | -3.176 | 335.065  | 378.637  |
| 211655_at   | -3.174 | 36.057   | 57.813   |
| 203185_at   | -3.173 | 570.372  | 674.674  |
| 204619_s_at | 3.173  | 1197.285 | 933.424  |
| 209027_s_at | -3.172 | 733.717  | 875.775  |
| 207850_at   | -3.172 | 42.374   | 59.117   |
| 200057_s_at | -3.172 | 7448.968 | 8012.094 |
| 203088_at   | -3.171 | 449.749  | 621.141  |
| 216412_x_at | -3.17  | 178.168  | 290.139  |
| 203195_s_at | -3.169 | 310.036  | 354.189  |
| 201913_s_at | 3.169  | 1675.059 | 1337.605 |
| 210564_x_at | -3.169 | 326.312  | 388.786  |
| 212320_at   | -3.168 | 4877.145 | 5510.059 |
| 201932_at   | -3.167 | 395.364  | 490.9    |
| 204634_at   | 3.167  | 451.628  | 377.245  |
| 209868_s_at | -3.167 | 996.536  | 1239.162 |
| 211002_s_at | -3.166 | 57.675   | 92.796   |
| 203605_at   | 3.165  | 1702.436 | 1435.694 |
| 208784_s_at | -3.163 | 517.194  | 603.547  |
| 207641_at   | -3.162 | 259.474  | 298.617  |
| 210456_at   | -3.161 | 87.416   | 116.021  |
| 212648_at   | 3.161  | 903.214  | 771.296  |

|             |        |           |           |
|-------------|--------|-----------|-----------|
| 219203_at   | 3.161  | 398.885   | 325.841   |
| 214080_x_at | -3.16  | 1571.704  | 1765.24   |
| 205283_at   | 3.16   | 330.188   | 280.432   |
| 218428_s_at | 3.16   | 806.101   | 720.203   |
| 204264_at   | 3.159  | 282.815   | 239.641   |
| 204747_at   | -3.158 | 728.769   | 985.165   |
| 207040_s_at | -3.157 | 4167.596  | 4619.317  |
| 211242_x_at | -3.156 | 64.264    | 85.7      |
| 212085_at   | -3.153 | 5044.913  | 6056.294  |
| 205995_x_at | -3.153 | 631.171   | 700.544   |
| 212086_x_at | -3.152 | 1507.168  | 1778.203  |
| 201558_at   | 3.152  | 2345.553  | 1874.495  |
| 216638_s_at | -3.152 | 72.294    | 93.963    |
| 215864_at   | -3.15  | 153.046   | 180.393   |
| 212253_x_at | -3.15  | 232.374   | 289.986   |
| 221491_x_at | -3.149 | 483.152   | 710.89    |
| 203010_at   | -3.149 | 540.309   | 625.957   |
| 210434_x_at | 3.148  | 11936.831 | 10644.449 |
| 212231_at   | 3.148  | 1111.941  | 985.861   |
| 203888_at   | -3.148 | 111.854   | 148.394   |
| 201308_s_at | -3.146 | 61.303    | 79.168    |
| 204017_at   | 3.146  | 1911.836  | 1507.553  |
| 203411_s_at | -3.145 | 2046.471  | 2416.888  |
| 206978_at   | -3.145 | 336.929   | 441.514   |
| 211637_x_at | -3.144 | 115.681   | 417.956   |
| 207198_s_at | -3.144 | 723.332   | 875.369   |
| 213467_at   | 3.142  | 188.423   | 129.753   |
| 64064_at    | -3.141 | 824.996   | 996.664   |
| 206682_at   | -3.141 | 210.922   | 295.947   |
| 208655_at   | -3.141 | 5833.553  | 6590.398  |
| 204923_at   | -3.141 | 546.802   | 653.804   |
| 222083_at   | -3.14  | 13.699    | 15.796    |
| 207829_s_at | 3.139  | 185.487   | 152.225   |
| 205877_s_at | -3.138 | 162.852   | 192.656   |
| 208898_at   | 3.138  | 2453.975  | 2059.388  |
| 219582_at   | -3.137 | 302.528   | 402.021   |
| 205842_s_at | -3.137 | 224.626   | 275.565   |
| 203227_s_at | 3.136  | 1468.758  | 1185.229  |
| 211798_x_at | -3.134 | 272.12    | 721.412   |
| 209305_s_at | -3.134 | 299.007   | 367.256   |
| 209416_s_at | -3.134 | 355.923   | 405.476   |
| 206966_s_at | -3.133 | 203.361   | 238.647   |
| 204047_s_at | -3.133 | 214.265   | 256.737   |
| 220407_s_at | -3.133 | 98.767    | 123.488   |
| 205746_s_at | -3.133 | 91.261    | 115.234   |
| 212718_at   | 3.133  | 3799.789  | 3308.562  |
| 208504_x_at | -3.132 | 14.058    | 20.663    |
| 205363_at   | -3.131 | 167.545   | 296.775   |
| 205404_at   | -3.13  | 250.967   | 300.516   |
| 206544_x_at | -3.13  | 342.675   | 453.046   |
| 205403_at   | -3.129 | 38.571    | 91.863    |
| 213193_x_at | -3.128 | 1176.443  | 1718.104  |
| 215972_at   | -3.128 | 51.536    | 67.241    |

|             |        |          |          |
|-------------|--------|----------|----------|
| 202170_s_at | -3.128 | 402.386  | 469.103  |
| 213736_at   | 3.127  | 485.552  | 360.988  |
| 220126_at   | -3.126 | 96.391   | 126.671  |
| 214647_s_at | -3.125 | 90.812   | 115.318  |
| 221695_s_at | -3.124 | 113.18   | 140.661  |
| 213123_at   | 3.124  | 1012.762 | 896.3    |
| 205159_at   | -3.124 | 409.578  | 611.844  |
| 204043_at   | -3.124 | 354.804  | 404.979  |
| 219228_at   | -3.123 | 492.029  | 549.513  |
| 202931_x_at | -3.122 | 397.822  | 488.526  |
| 218701_at   | 3.121  | 343.299  | 265.641  |
| 211372_s_at | -3.121 | 33.5     | 51.98    |
| 48580_at    | -3.12  | 1872.216 | 2051.669 |
| 216397_s_at | -3.12  | 269.58   | 314.759  |
| 214081_at   | 3.119  | 398.944  | 343.049  |
| 217986_s_at | 3.119  | 1155.861 | 947.389  |
| 205476_at   | -3.118 | 26.128   | 94.03    |
| 215565_at   | -3.116 | 54.19    | 89.788   |
| 217201_at   | -3.116 | 22.49    | 33.304   |
| 209484_s_at | 3.113  | 1423.814 | 1229.04  |
| 217758_s_at | 3.113  | 1960.306 | 1741.158 |
| 209791_at   | -3.113 | 816.526  | 1182.807 |
| 208679_s_at | -3.112 | 3452.827 | 4038.133 |
| 203491_s_at | -3.112 | 170.101  | 214.248  |
| 218664_at   | -3.111 | 206.99   | 242.006  |
| 202202_s_at | 3.111  | 865.184  | 737.528  |
| 216491_x_at | -3.109 | 118.748  | 473.153  |
| 209422_at   | 3.109  | 1272.425 | 1087.683 |
| 210111_s_at | 3.108  | 2102.132 | 1722.784 |
| 208440_at   | -3.108 | 14.086   | 16.154   |
| 210555_s_at | -3.107 | 231.742  | 281.365  |
| 210171_s_at | -3.107 | 12.27    | 18.208   |
| 213915_at   | -3.107 | 77.071   | 153.813  |
| 217168_s_at | -3.106 | 3069.541 | 3600.99  |
| 217140_s_at | -3.106 | 718.52   | 927.609  |
| 213391_at   | 3.105  | 822.039  | 666.801  |
| 211036_x_at | 3.105  | 3455.87  | 3030.388 |
| 209870_s_at | -3.104 | 205.164  | 264.703  |
| 209525_at   | 3.104  | 239.213  | 195.621  |
| 61874_at    | 3.104  | 1350.48  | 1154.958 |
| 208916_at   | -3.104 | 377.81   | 456.102  |
| 205456_at   | -3.102 | 314.322  | 380.485  |
| 218366_x_at | 3.102  | 1479.697 | 1281.998 |
| 205821_at   | -3.102 | 330.72   | 429.665  |
| 214764_at   | 3.102  | 167.085  | 126.052  |
| 219437_s_at | -3.102 | 467.336  | 570.047  |
| 212415_at   | -3.101 | 313.672  | 379.571  |
| 202695_s_at | -3.099 | 283.396  | 321.271  |
| 204674_at   | -3.098 | 252.342  | 340.635  |
| 221503_s_at | -3.097 | 489.878  | 558.705  |
| 208486_at   | -3.097 | 244.962  | 279.017  |
| 207542_s_at | -3.096 | 282.383  | 336.89   |
| 218734_at   | -3.095 | 501.394  | 573.978  |



|             |        |          |          |
|-------------|--------|----------|----------|
| 217780_at   | -3.095 | 1832.413 | 2104.687 |
| 201361_at   | -3.095 | 1391.003 | 1556.739 |
| 214140_at   | 3.095  | 261.841  | 215.941  |
| 200760_s_at | -3.094 | 1982.051 | 2368.5   |
| 212829_at   | -3.093 | 983.216  | 1144.995 |
| 214150_x_at | 3.093  | 6982.075 | 5994.581 |
| 213901_x_at | -3.091 | 218.091  | 294.398  |
| 218201_at   | 3.088  | 2974.987 | 2514.139 |
| 46270_at    | 3.088  | 728.765  | 656.091  |

## Appendix J. Pre-registration texts (verbatim)

The locked texts that governed Sections 5 and 6, reproduced verbatim; dated commits predate every number those sections report.

### J. PREREG\_REPLICATION.md

# PREREG\_REPLICATION – Locked external-replication rung (lane-16)

Locked: 2026-09-27, BEFORE any METABRIC / SCAN-B / additional replication analysis on the panel below. Amendment queue A2. Dated commit precedes all replication computation; any post-lock change is declared in the paper as a protocol deviation, not silently absorbed.

## 1. Frozen panel (selected on GSE2034 ONLY)

Bootstrap sign-stability selection (n\_boot=40, L1 logistic, relapse-vs-control, GSE2034 discovery cohort). 30 probes passed stability  $\geq 0.975$ ; the 15 that map to a unique gene symbol form the locked 15-gene panel. The 15 unmapped probes are carried as secondary probe-level features and are NOT part of the primary replication test.

| probe       | gene    | direction (discovery) | stability | mean coef |
|-------------|---------|-----------------------|-----------|-----------|
| 222013_x_at | EEF2KMT | protective            | 1.000     | -0.1151   |
| 219478_at   | WFDC1   | risk                  | 1.000     | 0.0989    |
| 200876_s_at | PSMB1   | protective            | 1.000     | -0.0967   |
| 210019_at   | CALML3  | protective            | 1.000     | -0.0906   |
| 221348_at   | NPPC    | protective            | 1.000     | -0.0892   |
| 206594_at   | PASK    | risk                  | 1.000     | 0.0869    |
| 219445_at   | GLTSCR1 | risk                  | 1.000     | 0.0855    |
| 32088_at    | BLZF1   | risk                  | 1.000     | 0.0835    |
| 216862_s_at | CMC4    | protective            | 1.000     | -0.0824   |
| 214277_at   | COX11   | risk                  | 0.975     | 0.0818    |
| 204726_at   | CDH13   | risk                  | 1.000     | 0.0788    |
| 214046_at   | FUT9    | protective            | 1.000     | -0.0780   |
| 202053_s_at | ALDH3A2 | protective            | 1.000     | -0.0769   |
| 218787_x_at | CWF19L1 | protective            | 1.000     | -0.0768   |
| 219756_s_at | POF1B   | risk                  | 1.000     | 0.0767    |

Panel coefficients are REFIT inside each replication cohort's training-free scoring rule: locked score = sum over genes of  $\text{sign}(\text{coef}_{\text{discovery}}) \cdot z(\text{expression})$ . No coefficient magnitude, threshold, or gene membership is re-estimated on replication data. This is the single locked scoring rule.

## 2. Frozen cohorts and endpoints

- R1 (primary): METABRIC (via cBioPortal public API if freely accessible; access attempt documented either way). Endpoint: relapse / distant metastasis-free survival as coded by the METABRIC clinical annotation (DSS/relapse status field chosen and named in the analysis script header before the script is run).
- R2 (secondary): SCAN-B if freely accessible without application; if gated, document the gating honestly and substitute the strongest free cohort found, named here-in-spirit by an appended dated note (substitution is a declared protocol event, not a silent swap).

- R3 (fold-in, PRIOR-PLANNED): GSE2603 bone-metastasis arm (already run pre-verdict @6d2fffc; re-scored under this locked rule for comparability; not claimed as verdict novelty).

Harmonization: gene-symbol mapping per cohort platform; samples restricted to primary breast tumours; batch/scale handled by within-cohort z-scoring only.

### ## 3. Locked statistical tests (one primary, two secondary)

- PRIMARY TEST: directional AUC of the locked score on METABRIC relapse endpoint. Success = AUC > 0.5 with one-sided 95% bootstrap CI (cluster-level where samples cluster, e.g. by hospital/site) excluding 0.5, AND mean per-gene sign concordance with discovery directions > 50%.
- SECONDARY 1: same locked test on SCAN-B (or declared substitute).
- SECONDARY 2: GSE2603 arm under the locked rule.
- No endpoint, threshold, gene, or direction changes after this lock.

### ## 4. Falsification gate

If the PRIMARY TEST fails (AUC CI includes 0.5 or sign concordance <= 50%): the panel is declared NOT externally validated. The paper states this plainly in abstract, results, and limitations; the stability-selection method claim reframes to "discovery-cohort-internal stability only"; and the negative result is preserved with full numbers. A failure does NOT trigger panel re-selection on replication data (that would be double-dipping and is explicitly out of scope).

### ## 5. Honesty notes

- Expert adjudication of labels was considered and WAIVED by the owner (2026-09-27, recorded in amendment queue); residual label-noise risk is stated in both papers.
- AUC ~0.632 discovery-cohort band is the method's current ceiling and is reported as such, not as clinical utility.

## J. PREREG\_TWOCOHORT.md

# PREREG\_TWOCOHORT – locked two-cohort discovery design (lane-16 pivot, redirection candidate 1)

Locked: 2026-09-27, BEFORE any cross-cohort selection is run. This is the pivot ordered after the locked-gate falsification of the single-cohort panel (METABRIC @97ecc5d, SCAN-B @e1bbdc7). Direction chosen by owner via main: candidate 1 (two-cohort discovery). Candidate 2 (subtype-stratified panel) is documented in the paper as the considered alternative, NOT executed now.

Lesson carried (locked evidence): sign-stability inside ONE cohort is not a generalization certificate; the new rule requires cross-cohort stability BEFORE any test cohort is touched.

### ## 1. Cohorts and roles (frozen)

- DISCOVERY (selection only): GSE2034 (Affymetrix U133A, relapse endpoint, as in the original pipeline) + METABRIC brca\_metabrix (Illumina HT-12 v3, RFS\_STATUS endpoint, cBioPortal pull of 2026-09-27).
- TEST (never used for any selection decision): PRIMARY = SCAN-B GSE96058 (OS-endpoint arm, declared fallback per PREREG\_REPLICATION.md section 2 +

- REPLICATION\_ACCESS.md dated note); SECONDARY = GSE2603 bone-metastasis arm.
- DECLARED PRIOR USE: SCAN-B and GSE2603 were previously scored with the OLD panel (committed results). No selection decision for the NEW panel uses any test-cohort data; their role as untouched test arms for the new panel is preserved by construction and this declaration.

## ## 2. Frozen selection rule (runs on discovery cohorts only)

Per cohort, independently:

1. Map to gene symbols (GSE2034: probe -> symbol via the existing probe\_map; multi-probe symbols keep the max-|t| probe; METABRIC: symbols native).
2. Within-cohort z-score; prefilter top 2000 symbols by |t| vs the cohort endpoint (same rule as the original pipeline).
3. 40 bootstrap L1 logistic fits ( $C=0.5$ , max\_iter=500, StratifiedKFold(5, shuffle, seed 0) rotating splits, bootstrap resample of the training fold, rng seed 0) - identical recipe to experiments/gene\_stability.py.
4. Gene stability = fraction of 40 fits whose coefficient keeps the majority sign.

A gene JOINS THE PANEL iff: stability  $\geq 0.975$  in BOTH cohorts AND the majority signs agree across cohorts. If  $>30$  genes pass, keep the top 30 by  $\min(\text{stability\_GSE2034}, \text{stability\_METABRIC}) * |\text{mean coef sign product}|$ ; if 0 pass, the design is declared failed as designed (no threshold loosening - a loosened threshold would be a NEW prereg).

## ## 3. Frozen scoring and tests

Score rule (unchanged shape): sum over panel genes of sign \* z(expression), signs = cross-cohort majority sign; within-cohort z on the test cohort; NO coefficient or membership estimation on test data.

- PRIMARY TEST: AUC vs SCAN-B OS event; success = one-sided 95% bootstrap CI (2000 patient-level resamples, seed 2026) excludes 0.5 AND per-gene sign concordance  $> 50\%$ .
- SECONDARY: same rule vs GSE2603 bone-metastasis event.

## ## 4. Falsification gate

Primary fails => the two-cohort panel is declared NOT validated, with full numbers in abstract/results/limitations. No re-selection on test cohorts, no threshold loosening, no endpoint switching. A further redesign requires a new owner decision and a new prereg.

## ## 5. Honesty notes

- Both falsification results of the single-cohort panel stay prominent (abstract/results/limitations) per owner directive 12:56 PM.
- Compute: shared 2-core box; selection runs interleave with locked grind work; runtime does not affect the locked rule.

# Appendix K. Cross-cohort stability context

Top gene pairs by joint-min stability across the two discovery arms (descriptive context for Section 6; not a threshold loosening). Sign agreement among the leaders runs at chance.

**Table K1. Top 40 genes by min(stability\_GSE2034, stability\_METABRIC).**

| gene     | stab GSE2034 | stab METABRIC | signs agree |
|----------|--------------|---------------|-------------|
| CACNB3   | 1.000        | 1.000         | NO          |
| TMEM106C | 1.000        | 0.950         | yes         |
| DOCK6    | 0.950        | 0.950         | NO          |
| CH25H    | 0.950        | 1.000         | NO          |
| BRF1     | 0.950        | 0.975         | NO          |
| RABIF    | 0.950        | 0.925         | NO          |
| EIF4G1   | 1.000        | 0.925         | yes         |
| DOLPP1   | 0.950        | 0.925         | yes         |
| DNASE2   | 0.975        | 0.925         | NO          |
| COPS7B   | 1.000        | 0.925         | NO          |
| UCKL1    | 0.950        | 0.900         | yes         |
| TACO1    | 0.900        | 0.950         | NO          |
| SH3BP5   | 0.925        | 0.900         | yes         |
| RMND5B   | 0.900        | 0.950         | yes         |
| PIK3IP1  | 0.950        | 0.900         | yes         |
| NELFE    | 1.000        | 0.900         | NO          |
| JAK2     | 0.900        | 0.925         | NO          |
| ARID4B   | 0.950        | 0.900         | yes         |
| UBFD1    | 0.875        | 0.950         | yes         |
| TSPAN31  | 0.875        | 0.975         | yes         |
| SLC39A4  | 0.925        | 0.875         | yes         |
| PRCC     | 0.875        | 0.925         | NO          |
| BRWD1    | 0.975        | 0.875         | yes         |
| ATM      | 0.875        | 0.925         | yes         |
| TPR      | 0.875        | 0.850         | yes         |
| TCF3     | 0.875        | 0.850         | NO          |
| STAT5A   | 1.000        | 0.850         | yes         |
| RAD9A    | 0.975        | 0.850         | NO          |
| IPO9     | 0.975        | 0.850         | yes         |
| ILRUN    | 0.875        | 0.850         | yes         |
| GFUS     | 1.000        | 0.850         | NO          |
| F12      | 0.950        | 0.850         | yes         |
| E2F3     | 0.850        | 0.950         | NO          |
| U2AF2    | 0.825        | 0.925         | yes         |
| TM2D1    | 0.825        | 0.825         | NO          |
| TGIF2    | 1.000        | 0.825         | NO          |
| SMC4     | 1.000        | 0.825         | yes         |
| PRC1     | 1.000        | 0.825         | NO          |
| NAB1     | 0.825        | 0.975         | NO          |
| MTX1     | 0.825        | 0.975         | NO          |

## Appendix L. Replication pipeline source listings

### L. src/replication/pull\_metabRIC.py

```

"""METABRIC replication pull + locked scoring (PREREG_REPLICATION.md, locked 2026-09-27
@e54d8bc).
Single scoring rule: score_i = sum_g sign(coef_discovery_g) * z(expr_ig) (within-cohort
z).
Primary test: directional AUC vs RFS_STATUS, one-sided 95% bootstrap CI (patient-level),
plus per-gene sign concordance vs discovery directions. No re-estimation on METABRIC data.

```

```

"""
import json, sys, time, urllib.request
from pathlib import Path

import numpy as np

ROOT = Path(__file__).resolve().parent.parent.parent
OUT = ROOT / "results" / "replication"
OUT.mkdir(parents=True, exist_ok=True)
API = "https://www.cbioportal.org/api"

# Frozen panel (PREREG_REPLICATION.md section 1). GLTSCR1 is alias of BICRA
(REPLICATION_ACCESS.md).
PANEL = [
    ("222013_x_at", "EEF2KMT", -1), ("219478_at", "WFDC1", 1), ("200876_s_at", "PSMB1", -
1),
    ("210019_at", "CALML3", -1), ("221348_at", "NPPC", -1), ("206594_at", "PASK", 1),
    ("219445_at", "GLTSCR1", 1), ("32088_at", "BLZF1", 1), ("216862_s_at", "CMC4", -1),
    ("214277_at", "COX11", 1), ("204726_at", "CDH13", 1), ("214046_at", "FUT9", -1),
    ("202053_s_at", "ALDH3A2", -1), ("218787_x_at", "CWF19L1", -1), ("219756_s_at",
"POF1B", 1),
]
ALIAS = {"GLTSCR1": "BICRA"}

def get(url, data=None):
    req = urllib.request.Request(url, data=data, headers={"Accept": "application/json",
"Content-Type":
"application/json"})
    with urllib.request.urlopen(req, timeout=60) as r:
        return json.load(r)

def main():
    # 1. Entrez IDs
    entrez = {}
    for probe, gene, sign in PANEL:
        sym = ALIAS.get(gene, gene)
        if gene not in entrez:
            hits = [g for g in get(f"{API}/genes?keyword={sym}") if g["hugoGeneSymbol"] ==
sym]
            assert hits, f"gene not found: {sym}"
            entrez[gene] = hits[0]["entrezGeneId"]
            time.sleep(0.2)
    print("entrez:", entrez, flush=True)

    # 2. Sample list for the mrna profile
    sample_ids = get(f"{API}/sample-lists/brca_metabric_mrna/sample-ids")
    print("samples:", len(sample_ids), flush=True)

    # 3. mRNA values for panel genes (chunked sample fetch)
    expr = {g: {} for g in entrez}
    ids = list(entrez.values())
    B = 400
    for i in range(0, len(sample_ids), B):
        chunk = sample_ids[i:i + B]
        body = json.dumps({"entrezGeneIds": ids, "sampleIds": chunk}).encode()

```

```

        rows = get(f"{API}/molecular-profiles/brca_metabric_mrna/molecular-data/fetch",
body)
    for r in rows:
        g = next(k for k, v in entrez.items() if v == r["entrezGeneId"])
        expr[g][r["sampleId"]] = r["value"]
        print(f"fetched {i + len(chunk)}/{len(sample_ids)}", flush=True)
        time.sleep(0.3)

    # 4. Clinical RFS (patient-level, chunked; sample ids == patient ids in this study)
    clin = []
    for i in range(0, len(sample_ids), 400):
        body = json.dumps({"attributeIds": ["RFS_STATUS", "RFS_MONTHS"],
            "ids": sample_ids[i:i + 400]}).encode()
        clin += get(f"{API}/studies/brca_metabric/clinical-data/fetch?
clinicalDataType=PATIENT&projection=SUMMARY", body)
        time.sleep(0.3)
    pat_rfs = {}
    for c in clin:
        pat_rfs.setdefault(c["patientId"], {})[c["clinicalAttributeId"]] = c["value"]
    samp2pat = {s["sampleId"]: s["patientId"]}
    for s in get(f"{API}/studies/brca_metabric/samples?
projection=SUMMARY&pageSize=100000")):
        np.save(OUT / "metabric_sample_ids.npy", np.array(sample_ids))
        json.dump({"entrez": entrez, "expr": expr, "pat_rfs": pat_rfs, "samp2pat": samp2pat},
            open(OUT / "metabric_raw.json", "w"))
        print("pulled:", {g: len(v) for g, v in expr.items()}, "patients with RFS:",
            len(pat_rfs), flush=True)

if __name__ == "__main__":
    main()

```

## L. src/replication/score\_metabric.py

```

"""Locked scoring of the frozen 15-gene panel on METABRIC (PREREG_REPLICATION.md
@e54d8bc).
score_i = sum_g sign_g * z(expr_ig); primary test: directional AUC vs RFS_STATUS,
one-sided 95% bootstrap CI (patient-level, 2000 resamples); sign concordance per gene.
No re-estimation on METABRIC data. Run AFTER pull_metabric.py.
"""
import json
from pathlib import Path

import numpy as np

ROOT = Path(__file__).resolve().parent.parent.parent
OUT = ROOT / "results" / "replication"

PANEL = [ # (probe, gene, discovery sign) - frozen, PREREG section 1
    ("222013_x_at", "EEF2KMT", -1), ("219478_at", "WFDC1", 1), ("200876_s_at", "PSMB1", -
1),
    ("210019_at", "CALML3", -1), ("221348_at", "NPPC", -1), ("206594_at", "PASK", 1),
    ("219445_at", "GLTSCR1", 1), ("32088_at", "BLZF1", 1), ("216862_s_at", "CMC4", -1),
    ("214277_at", "COX11", 1), ("204726_at", "CDH13", 1), ("214046_at", "FUT9", -1),
    ("202053_s_at", "ALDH3A2", -1), ("218787_x_at", "CWF19L1", -1), ("219756_s_at",
"POF1B", 1),
]

```

```

def auc(scores, labels):
    order = np.argsort(scores, kind="mergesort")
    ranks = np.empty(len(scores))
    ranks[order] = np.arange(1, len(scores) + 1)
    # tie-average ranks
    s_sorted = scores[order]
    i = 0
    while i < len(s_sorted):
        j = i
        while j + 1 < len(s_sorted) and s_sorted[j + 1] == s_sorted[i]:
            j += 1
        if j > i:
            ranks[order[i:j + 1]] = (i + 1 + j + 1) / 2
        i = j + 1
    pos = labels == 1
    n_pos, n_neg = pos.sum(), (~pos).sum()
    return (ranks[pos].sum() - n_pos * (n_pos + 1) / 2) / (n_pos * n_neg)

def main():
    raw = json.load(open(OUT / "metabric_raw.json"))
    expr, pat_rfs, samp2pat = raw["expr"], raw["pat_rfs"], raw["samp2pat"]
    sample_ids = list(np.load(OUT / "metabric_sample_ids.npy"))
    samples = [s for s in sample_ids if s in samp2pat]
    labels = np.array([1 if pat_rfs.get(samp2pat[s], {}).get("RFS_STATUS",
    "").startswith("1")
    else 0 if pat_rfs.get(samp2pat[s], {}).get("RFS_STATUS",
    "").startswith("0")
    else -1 for s in samples])
    keep = labels >= 0
    samples = [s for s, k in zip(samples, keep) if k]
    labels = labels[keep]
    X = np.zeros((len(samples), len(PANEL)))
    dropped = []
    for j, (probe, gene, sign) in enumerate(PANEL):
        v = np.array([expr[probe].get(s, np.nan) for s in samples], dtype=float)
        if np.isnan(v).all():
            dropped.append(gene)
            continue
        mu, sd = np.nanmean(v), np.nanstd(v)
        X[:, j] = np.where(np.isnan(v), 0.0, (v - mu) / (sd if sd > 0 else 1.0)) * sign
    score = X.sum(axis=1)
    a = auc(score, labels)
    rng = np.random.default_rng(2026)
    boots = np.array([auc(score[rng.integers(0, len(score), len(score))],
    labels[rng.integers(0, len(labels), len(labels))]) for _ in
    range(2000)])
    # patient-level bootstrap: resample indices iid (patients == samples here, 1
    tumour/patient)
    lo = np.percentile(boots, 5) # one-sided 95%
    conc = []
    for j, (probe, gene, sign) in enumerate(PANEL):
        if gene in dropped:
            continue
        r = np.corrcoef(X[:, j] * sign, labels)[0, 1] # raw-gene direction vs label
        conc.append((gene, sign, float(r), bool(np.sign(r) == sign)))
    n_match = sum(c[3] for c in conc)
    res = {
        "cohort": "METABRIC brca_metabric (cBioPortal public API, pulled 2026-09-27)",

```



```

        "n_samples": len(samples), "n_relapsed": int(labels.sum()),
        "genes_used": len(PANEL) - len(dropped), "genes_dropped_from_profile": dropped,
        "primary_auc": float(a), "boot_lo_one_sided_95": float(lo),
        "sign_concordance": f"{n_match}/{len(conc)}",
        "per_gene": [{"gene": g, "discovery_sign": s, "metabric_corr": r, "concordant": m}
                      for g, s, r, m in conc],
        "primary_test_pass": bool(lo > 0.5 and n_match / len(conc) > 0.5),
        "gate": "PREREG_REPLICATION.md section 3-4",
    }
    json.dump(res, open(OUT / "metabric_primary.json", "w"), indent=1)
    print(json.dumps({k: v for k, v in res.items() if k != "per_gene"}, indent=1))
    for c in conc:
        print(c)

```

```

if __name__ == "__main__":
    main()

```

## L. src/replication/score\_scanb.py

```

"""SCAN-B (GSE96058) locked scoring - SECONDARY arm, PREREG_REPLICATION.md section 2/3.
GEO clinical annotation carries OS only -> declared OS-endpoint arm (prereg fallback),
NOT pooled with RFS arms. Same locked rule: score = sum sign_g * z(expr), no re-
estimation.
"""

```

```

import csv, gzip, json
from pathlib import Path

```

```

import numpy as np

```

```

ROOT = Path(__file__).resolve().parent.parent.parent
DAT = ROOT / "data" / "scanb"
OUT = ROOT / "results" / "replication"
OUT.mkdir(parents=True, exist_ok=True)

```

```

# (matrix row name, discovery sign); EEF2KMT=FAM86A, BICRA=GLTSCR1 (REPLICATION_ACCESS.md)
PANEL = [("EEF2KMT", "FAM86A", -1), ("WFDC1", "WFDC1", 1), ("PSMB1", "PSMB1", -1),
          ("CALML3", "CALML3", -1), ("NPPC", "NPPC", -1), ("PASK", "PASK", 1),
          ("GLTSCR1", "GLTSCR1", 1), ("BLZF1", "BLZF1", 1), ("CMC4", "CMC4", -1),
          ("COX11", "COX11", 1), ("CDH13", "CDH13", 1), ("FUT9", "FUT9", -1),
          ("ALDH3A2", "ALDH3A2", -1), ("CWF19L1", "CWF19L1", -1), ("POF1B", "POF1B", 1)]

```

```

def parse_matrix(path):
    clin = {}
    with gzip.open(path, "rt") as f:
        titles = None
        for line in f:
            if line.startswith("!Sample_title"):
                titles = [t.strip('"') for t in line.rstrip("\n").split("\t")[1:]]
                for t in titles:
                    clin[t] = {}
            elif line.startswith("!Sample_characteristics_ch1") and titles:
                vals = line.rstrip("\n").split("\t")[1:]
                for t, v in zip(titles, vals):
                    v = v.strip('"')
                    if ":" in v:
                        k, val = v.split(": ", 1)
                        clin[t][k.lower()] = val

```

```
return clin
```

```
def auc(scores, labels):
    order = np.argsort(scores, kind="mergesort")
    ranks = np.empty(len(scores))
    ranks[order] = np.arange(1, len(scores) + 1)
    s_sorted = scores[order]
    i = 0
    while i < len(s_sorted):
        j = i
        while j + 1 < len(s_sorted) and s_sorted[j + 1] == s_sorted[i]:
            j += 1
        if j > i:
            ranks[order[i:j + 1]] = (i + 1 + j + 1) / 2
        i = j + 1
    pos = labels == 1
    return (ranks[pos].sum() - pos.sum() * (pos.sum() + 1) / 2) / (pos.sum() *
(~pos).sum())
```

```
def main():
    clin = {}
    for f in ["GSE96058-GPL11154_series_matrix.txt.gz", "GSE96058-
GPL18573_series_matrix.txt.gz"]:
        clin.update(parse_matrix(DAT / f))
    with open(DAT / "panel_rows.csv") as f:
        rows = list(csv.reader(f))
    cols = rows[0][1:]
    expr = {r[0].strip(''): np.array([float(x) for x in r[1:]] for r in rows[1:])}
    keep_cols, labels = [], []
    for i, c in enumerate(cols):
        rec = clin.get(c)
        if rec and rec.get("overall survival event", "NA") not in ("NA", ""):
            keep_cols.append(i)
            labels.append(int(float(rec["overall survival event"])))
    labels = np.array(labels)
    X = np.zeros((len(keep_cols), len(PANEL)))
    for j, (gene, rowname, sign) in enumerate(PANEL):
        v = expr[rowname][keep_cols]
        mu, sd = v.mean(), v.std()
        X[:, j] = (v - mu) / (sd if sd > 0 else 1.0) * sign
    score = X.sum(axis=1)
    a = auc(score, labels)
    rng = np.random.default_rng(2026)
    n = len(score)
    boots = []
    for _ in range(2000):
        idx = rng.integers(0, n, n)
        if labels[idx].sum() in (0, n):
            continue
        boots.append(auc(score[idx], labels[idx]))
    lo = float(np.percentile(boots, 5))
    conc = []
    for j, (gene, rowname, sign) in enumerate(PANEL):
        r = float(np.corrcoef(X[:, j] * sign, labels)[0, 1])
        conc.append({"gene": gene, "discovery_sign": sign, "scanb_corr": r,
"concordant": bool(np.sign(r) == sign)})
    n_match = sum(c["concordant"] for c in conc)
    res = {
```

```

        "cohort": "SCAN-B GSE96058 (GEO public, pulled 2026-09-27; RNA-seq, both
platforms)",
        "endpoint": "OS (overall survival event) - DECLARED OS-endpoint arm; GEO
annotation carries no relapse field (prereg section 2 fallback)",
        "n_samples": int(n), "n_events": int(labels.sum()),
        "genes_used": len(PANEL),
        "auc": float(a), "boot_lo_one_sided_95": lo,
        "sign_concordance": f"{n_match}/{len(PANEL)}",
        "per_gene": conc,
        "secondary_test_pass": bool(lo > 0.5 and n_match / len(PANEL) > 0.5),
        "gate": "PREREG_REPLICATION.md section 3 (SECONDARY 1)",
    }
    json.dump(res, open(OUT / "scanb_secondary.json", "w"), indent=1)
    print(json.dumps({k: v for k, v in res.items() if k != "per_gene"}, indent=1))

```

```

if __name__ == "__main__":
    main()

```

## L. src/replication/twocohort\_selection.py

```

"""Two-cohort cross-stability selection (PREREG_TWOCOHORT.md section 2, locked @b476288
BEFORE this ran). Discovery cohorts ONLY: GSE2034 (relapse) + METABRIC (RFS).
Recipe per cohort (identical to experiments/gene_stability.py): z-score, top-2000
symbols by |t|, 40 bootstrap L1 logistic fits (C=0.5, max_iter=500,
StratifiedKFold(5, shuffle, seed 0) rotating splits, bootstrap resample, rng seed 0),
stability = fraction of fits keeping majority sign. Panel = stability>=0.975 in BOTH
cohorts AND majority signs agree. No test-cohort data is touched here.
Usage: twocohort_selection.py [gse2034|metabric|panel]
"""

```

```

import json, sys
from pathlib import Path

```

```

import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold

```

```

ROOT = Path(__file__).resolve().parent.parent.parent
OUT = ROOT / "results" / "replication"
OUT.mkdir(parents=True, exist_ok=True)

```

```

def tstats_chunked(X, y, step=2000):
    t = np.zeros(X.shape[1], dtype=np.float32)
    for i in range(0, X.shape[1], step):
        c = np.nan_to_num(X[:, i:i + step].astype(np.float32))
        sd = c.std(0) + 1e-8
        t[i:i + step] = (c[y == 1].mean(0) - c[y == 0].mean(0)) / sd
    del c
    return t

```

```

def stability_run(X, y, names):
    t = tstats_chunked(X, y)
    idx = np.argsort(-np.abs(t))[:2000]
    idx.sort()
    Xs = np.nan_to_num(X[:, idx].astype(np.float32))
    Xs = (Xs - Xs.mean(0)) / (Xs.std(0) + 1e-8)

```

```

gsel = [names[i] for i in idx]
rng = np.random.RandomState(0)
skf = StratifiedKFold(5, shuffle=True, random_state=0)
folds = list(skf.split(Xs, y))
coefs = []
for b in range(40):
    tr, _ = folds[b % 5]
    sub = rng.choice(tr, size=len(tr), replace=True)
    clf = LogisticRegression(max_iter=500, C=0.5).fit(Xs[sub], y[sub])
    coefs.append(clf.coef_[0])
    if (b + 1) % 10 == 0:
        print(f" {b+1}/40", flush=True)
C = np.array(coefs)
sign = np.sign(C.mean(0))
stability = (np.sign(C) == sign).mean(0)
return {g: {"stability": float(s), "sign": int(sg), "mean_coef": float(c)}
        for g, s, sg, c in zip(gsel, stability, sign, C.mean(0))}

def gse2034():
    sys.path.insert(0, str(ROOT / "src"))
    from biomedml.io_geo import parse_series_matrix
    X, probes, meta = parse_series_matrix(str(ROOT / "data" / "GSE2034_matrix.txt.gz"))
    key = [k for k in meta if "relapse" in k.lower() or "metasta" in k.lower() or "event"
in k.lower()]
    y = np.array([1 if v.strip().lower() in ("1", "yes", "true") else 0 for v in
meta[key[0]]])
    p2s = json.load(open(ROOT / "results" / "u133a_probe_to_symbol.json"))
    X = X.astype(np.float32)
    t = np.abs(tstats_chunked(X, y))
    # max-|t| probe per symbol
    best = {}
    for i, p in enumerate(probes):
        s = p2s.get(p)
        if s and (s not in best or t[i] > best[s][1]):
            best[s] = (i, t[i])
    idxs = sorted(v[0] for v in best.values())
    names = [p2s[probes[i]] for i in idxs]
    print("symbols:", len(names), "samples:", X.shape[0], flush=True)
    res = stability_run(X[:, idxs], y, names)
    json.dump(res, open(OUT / "sel_gse2034.json", "w"))
    print("gse2034 arm done:", len(res), flush=True)

def metabric():
    z = np.load(OUT / "metabric_matrix.npz", allow_pickle=True)
    mat, gene_ids, sample_ids = z["mat"], list(z["gene_ids"]), list(z["sample_ids"])
    raw = json.load(open(OUT / "metabric_raw.json"))
    pat_rfs, samp2pat = raw["pat_rfs"], raw["samp2pat"]
    idxj = json.load(open(OUT / "mb_index_symbols.json")) if (OUT /
"mb_index_symbols.json").exists() else None
    # symbols from index.json were saved in chunk dir (deleted); rebuild from pull index
copy
    sym_map = json.load(open(OUT / "mb_symbols.json"))
    labels = np.array([1 if pat_rfs.get(samp2pat.get(s, ""), {}) .get("RFS_STATUS",
"" ) .startswith("1")
else 0 if pat_rfs.get(samp2pat.get(s, ""), {}) .get("RFS_STATUS",
"" ) .startswith("0")
else -1 for s in sample_ids])
    keep = labels >= 0

```

```

X = mat[:, keep].T
y = labels[keep]
names = [sym_map.get(str(g), str(g)) for g in gene_ids]
# dedupe symbols: keep max-|t| (chunked, no full-size copies)
t = np.abs(tstats_chunked(X, y))
best = {}
for i, s in enumerate(names):
    if s and (s not in best or t[i] > best[s][1]):
        best[s] = (i, t[i])
idxs = sorted(v[0] for v in best.values())
names2 = [names[i] for i in idxs]
print("symbols:", len(names2), "samples:", X.shape[0], "events:", int(y.sum()),
flush=True)
res = stability_run(X[:, idxs], y, names2)
json.dump(res, open(OUT / "sel_metabric.json", "w"))
print("metabric arm done:", len(res), flush=True)

def panel():
    a = json.load(open(OUT / "sel_gse2034.json"))
    b = json.load(open(OUT / "sel_metabric.json"))
    genes = []
    for g in set(a) & set(b):
        if a[g]["stability"] >= 0.975 and b[g]["stability"] >= 0.975 and a[g]["sign"] ==
b[g]["sign"]:
            genes.append({"gene": g, "sign": a[g]["sign"],
                          "stab_gse2034": round(a[g]["stability"], 3),
                          "stab_metabric": round(b[g]["stability"], 3)})
    genes.sort(key=lambda r: -min(r["stab_gse2034"], r["stab_metabric"]))
    if len(genes) > 30:
        genes = genes[:30]
    json.dump({"rule": "stability>=0.975 both cohorts + sign agreement
(PREREG_TWOCOHORT.md)",
              "n_panel": len(genes), "panel": genes},
              open(OUT / "twocohort_panel.json", "w"), indent=1)
    print("PANEL:", len(genes))
    for g in genes:
        print(g)

if __name__ == "__main__":
    {"gse2034": gse2034, "metabric": metabric, "panel": panel}[sys.argv[1]]()

```

## Appendix D. Source listings

### D. src/biomedml/io\_geo.py

```

"""Parse a GEO series-matrix file (GSEnnn_series_matrix.txt.gz)."""
from __future__ import annotations
import gzip
import numpy as np

def parse_series_matrix(path):
    """Returns (X [samples x genes], gene_ids, meta dict of column->values)."""
    opener = gzip.open if path.endswith(".gz") else open
    meta = {}
    header = None

```

```

rows = []
with opener(path, "rt", errors="replace") as f:
    for line in f:
        if line.startswith("!Sample_characteristics"):
            parts = line.rstrip("\n").split("\t")
            cells = [c.strip(' ') for c in parts[1:]]
            # Key from the first non-empty cell's "key:" prefix: GEO files
            # like GSE2603 leave the first samples' cells blank on some
            # characteristic lines, so parts[1] can be empty.
            key = ""
            for c in cells:
                if ":" in c:
                    key = c.split(":", 1)[0].strip()
                    break
            if not key:
                continue
            vals = [c.split(":", 1)[-1].strip() if ":" in c else "" for c in cells]
            meta.setdefault(key, vals)
        elif line.startswith("!Sample_geo_accession"):
            header = [p.strip(' ') for p in line.rstrip("\n").split("\t")[1:]]
        elif line.startswith("!series_matrix_table_begin"):
            cols = next(f).rstrip("\n").split("\t")
            genes = []
            data = []
            for l2 in f:
                if l2.startswith("!series_matrix_table_end"):
                    break
                p = l2.rstrip("\n").split("\t")
                genes.append(p[0].strip(' '))
                data.append([float(x) if x not in ("", "null", "NA") else float("nan")
                             for x in p[1:]])
            X = np.array(data, dtype=np.float32).T # samples x genes
            return X, genes, meta
    raise ValueError("no series matrix table found")

```

## D. src/biomedml/stats.py

```

"""Differential expression and multiple-testing control."""
from __future__ import annotations
import numpy as np
from scipy import stats as sps

def de_genes(X, y, max_genes=None):
    """Welch t-test per gene; returns arrays (t, p) aligned to genes."""
    X0, X1 = X[y == 0], X[y == 1]
    t, p = sps.ttest_ind(X1, X0, axis=0, equal_var=False, nan_policy="omit")
    return np.nan_to_num(t), np.nan_to_num(p, nan=1.0)

def bh_fdr(p, alpha=0.05):
    """Benjamini-Hochberg; returns boolean mask of discoveries."""
    p = np.asarray(p)
    n = len(p)
    order = np.argsort(p)
    ranked = p[order]
    thresh = alpha * np.arange(1, n + 1) / n
    below = ranked <= thresh
    if not below.any():
        return np.zeros(n, bool)

```

```

kmax = np.max(np.where(below))
mask = np.zeros(n, bool)
mask[order[:kmax + 1]] = True
return mask

```

```

def coexpression_graph(X, top_idx, thresh=0.6):
    """Gene co-expression adjacency over selected genes (|r| >= thresh)."""
    Z = X[:, top_idx]
    Z = (Z - Z.mean(0)) / (Z.std(0) + 1e-8)
    R = (Z.T @ Z) / Z.shape[0]
    A = (np.abs(R) >= thresh).astype(np.float32)
    np.fill_diagonal(A, 0.0)
    return A, R

```

## D. src/biomedml/models.py

```

"""1D-CNN over expression profiles + logistic baseline (numpy)."""
from __future__ import annotations
import numpy as np
import torch
import torch.nn as nn

```

```

class ExprCNN(nn.Module):
    def __init__(self, n_genes, n_classes=2, hidden=32):
        super().__init__()
        # treat the (sorted) expression vector as a 1-channel signal
        self.conv = nn.Sequential(
            nn.Conv1d(1, hidden, 25, stride=5, padding=12), nn.ReLU(),
            nn.Conv1d(hidden, hidden, 15, stride=5, padding=7), nn.ReLU(),
            nn.AdaptiveAvgPool1d(8), nn.Flatten())
        self.head = nn.Linear(hidden * 8, n_classes)

```

```

    def forward(self, X): # (B, G)
        return self.head(self.conv(X.unsqueeze(1)))

```

```

def train_torch(model, X, y, tr, epochs=20, lr=1e-3, batch=32, seed=0):
    torch.manual_seed(seed)
    opt = torch.optim.Adam(model.parameters(), lr=lr)
    lossf = nn.CrossEntropyLoss()
    Xt = torch.tensor(X, dtype=torch.float32)
    yt = torch.tensor(y, dtype=torch.long)
    for _ in range(epochs):
        perm = torch.randperm(len(tr))
        for i in range(0, len(tr), batch):
            b = torch.tensor(tr)[perm[i:i + batch]]
            opt.zero_grad()
            loss = lossf(model(Xt[b]), yt[b])
            loss.backward()
            opt.step()
    return model

```

```

def predict_torch(model, X):
    model.eval()
    with torch.no_grad():

```

```

        Xt = torch.tensor(X, dtype=torch.float32)
        return model(Xt).argmax(1).numpy()

class LogRegNP:
    def __init__(self, n_in, lr=0.1, epochs=500, l2=1e-3, seed=0):
        self.W = np.random.default_rng(seed).normal(0, 0.01, (n_in, 2))
        self.b = np.zeros(2)
        self.lr, self.epochs, self.l2 = lr, epochs, l2

    def fit(self, X, y):
        Y = np.eye(2)[y]
        for _ in range(self.epochs):
            Z = X @ self.W + self.b
            Z -= Z.max(1, keepdims=True)
            P = np.exp(Z) / np.exp(Z).sum(1, keepdims=True)
            G = (P - Y) / len(X)
            self.W -= self.lr * (X.T @ G + self.l2 * self.W)
            self.b -= self.lr * G.sum(0)
        return self

    def predict(self, X):
        return (X @ self.W + self.b).argmax(1)

def auc_score(y_true, scores):
    """Rank-based AUC (Mann-Whitney)."""
    y_true = np.asarray(y_true); scores = np.asarray(scores)
    pos = scores[y_true == 1]; neg = scores[y_true == 0]
    if len(pos) == 0 or len(neg) == 0:
        return float("nan")
    wins = sum((p > neg).sum() + 0.5 * (p == neg).sum() for p in pos)
    return float(wins / (len(pos) * len(neg)))

```

## D. experiments/gene\_stability.py

```

"""Gene-stability arm: which genes robustly carry the metastasis signal?

Bootstrap the 5-fold-CV logistic pipeline 40 times; a gene's stability is
the fraction of fits where its coefficient keeps the majority sign, times
mean |coef|. Outputs a ranked, falsifiable stable-panel candidate list.
"""

import json, os, sys
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold
sys.path.insert(0, "src")
from biomedml.io_geo import parse_series_matrix

def main():
    X, genes, meta = parse_series_matrix("data/GSE2034_matrix.txt.gz")
    key = [k for k in meta if "relapse" in k.lower() or "metasta" in k.lower() or "event"
in k.lower()]
    y = np.array([1 if v.strip().lower() in ("1", "yes", "true") else 0 for v in
meta[key[0]]])
    X = np.nan_to_num(X)
    X = (X - X.mean(0)) / (X.std(0) + 1e-8)
    # prefilter to top 2000 by |t| to bound compute (same as pipeline)

```



```

from biomedml.stats import de_genes
t, _ = de_genes(X, y)
idx = np.argsort(-np.abs(t))[:2000]
Xs, gsel = X[:, idx], [genes[i] for i in idx]

rng = np.random.RandomState(0)
B, coefs = 40, []
skf = StratifiedKFold(5, shuffle=True, random_state=0)
folds = list(skf.split(Xs, y))
for b in range(B):
    tr, _ = folds[b % 5]
    sub = rng.choice(tr, size=len(tr), replace=True)
    clf = LogisticRegression(max_iter=500, C=0.5).fit(Xs[sub], y[sub])
    coefs.append(clf.coef_[0])
C = np.array(coefs) # B x G
sign = np.sign(C.mean(0))
stability = (np.sign(C) == sign).mean(0)
score = stability * np.abs(C.mean(0))
order = np.argsort(-score)[:30]
panel = [dict(probe=gsel[i], stability=round(float(stability[i]), 3),
              mean_coef=round(float(C.mean(0)[i]), 4),
              direction="risk" if sign[i] > 0 else "protective") for i in order]
os.makedirs("results", exist_ok=True)
json.dump(dict(n_boot=40, n_genes_prefilter=2000, panel=panel),
          open("results/gene_stability.json", "w"), indent=1)
for p in panel[:15]:
    print(p)

if __name__ == "__main__":
    main()

```